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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Mechanical Properties of Fluids

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	151 questions
Numerical	46 questions
Answer key	at the end

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MCQ (Single Correct Answer) 151 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A liquid of density $600\ \mathrm{kg/m^3}$ flowing steadily in a tube of varying cross-section. The cross-section at a point A is $1.0\ \mathrm{cm^2}$ and that at B is $20\ \mathrm{mm^2}$. Both the points A and B are in same horizontal plane, the speed of the liquid at A is $10\ \mathrm{cm/s}$. The difference in pressures at A and B points is ____ Pa.

Open solution

(A) 18 (B) 144 (C) 36 (D) 72

Q2. JEE Main 2026 (Online) 8th April Evening Shift

A spherical liquid drop of radius R acquires the terminal velocity v_1 when falls through a gas of viscosity η. Now the drop is broken into 64 identical droplets and each droplet acquires terminal velocity v_2 falling through the same gas. The ratio of terminal velocities v_1/v_2 is ____ .

Open solution

(A) 4 (B) 0.25 (C) 32 (D) 16

Q3. JEE Main 2026 (Online) 6th April Evening Shift

A cylindrical vessel of 40 cm radius is completely filled with water and its capacity is $528\mathrm{dm^3}$ (dm : decimeter) The vessel is placed on a solid block of exactly same height as vessel. If a small hole is made at 70 cm below the top of water level, then horizontal range of water falling on the ground in the beginning is ____ cm .

Open solution

(A) $120\sqrt{2}$ (B) $140\sqrt{2}$ (C) $140\sqrt{3}$ (D) $120\sqrt{3}$

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Eight mercury drops, each of radius r, coalesce to form a bigger drop. The surface energy released in this process is ____ - (S is the surface tension of mercury).

Open solution

(A) $8\pi r^2\ \mathrm{S}$ (B) $16\pi r^2 S$ (C) $64\pi r^2 S$ (D) $4\pi r^2\ \mathrm{S}$

Q5. JEE Main 2026 (Online) 4th April Evening Shift

A water spray gun is attached to a hose of cross sectional area $30\ \mathrm{cm^2}$. The gun comprises of 10 perforations each of cross sectional area of $15\ \mathrm{mm^2}$. If the water flows in the hose with the speed of $50\ \mathrm{cm/s}$, calculate the speed at which the water flows out from each perforation. (Neglect any edge effects)

Open solution

(A) $100\ \mathrm{m/s}$ (B) $10\ \mathrm{m/s}$
(C) $1000\ \mathrm{m/s}$ (D) $15 \times 10^2\ \mathrm{m/s}$

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Q6. JEE Main 2026 (Online) 2nd April Evening Shift

The surface tension of a soap bubble is 0.03 N/m. The work done in increasing the diameter of bubble from 2 cm to 6 cm is $\alpha\pi \times 10^{-4}$ J. The value of α is _____. (Take $\pi = 3.14$)

(A) 0.86 (B) 0.64 (C) 1.92 (D) 7.68

Open solution

Q7. JEE Main 2026 (Online) 2nd April Evening Shift

If an air bubble of diameter 2 mm rises steadily through a liquid of density 2000 kg/m^3 at a rate of 0.5 cm/s, then the coefficient of viscosity of liquid is _____ Poise. (Take g = 10 m/s^2)

(A) 0.88 (B) 8.8 (C) 88.8 (D) 0.088

Open solution

Q8. JEE Main 2026 (Online) 2nd April Morning Shift

A liquid drop of diameter 2 mm breaks into 512 droplets. The change in surface energy is $\alpha \times 10^{-6}$ J. The value of α is _____. (Take surface tension of liquid = 0.08 N/m)

(A) 10 (B) 7 (C) 8 (D) 11

Open solution

Q9. JEE Main 2026 (Online) 24th January Evening Shift

A cubical block of density $\rho_b = 600\ \mathrm{kg/m^3}$ floats in a liquid of density $\rho_e = 900\mathrm{kg/m^3}$. If the height of block is $H = 8.0\ \mathrm{cm}$ then height of the submerged part is

____ cm .

(A) 6.3 (B) 4.3 (C) 7.3 (D) 5.3

Open solution

Q10. JEE Main 2026 (Online) 23rd January Evening Shift

A small metallic sphere of diameter 2 mm and density $10.5\ \mathrm{g/cm^3}$ is dropped in glycerine having viscosity 10 Poise and density $1.5\ \mathrm{g/cm^3}$ respectively. The terminal velocity attained by the sphere is ____ $\mathrm{cm/s}$.

$\left(\pi = \frac{22}{7}\right.$ and $\left.g = 10\ \mathrm{m/s^2}\right)$

(A) 1.0 (B) 1.5 (C) 3.0 (D) 2.0

Open solution

Q11. JEE Main 2026 (Online) 23rd January Evening Shift

An air bubble of volume $2.9\ \mathrm{cm^3}$ rises from the bottom of a swimming pool of 5 m deep. At the bottom of the pool water temperature is $17^\circ\mathrm{C}$. The volume of the bubble when it reaches the surface, where the water temperature is $27^\circ\mathrm{C}$, is ____ $\mathrm{cm^3}$.

($\mathrm{g} = 10\ \mathrm{m/s^2}$, density of water $= 10^3\ \mathrm{kg/m^3}$, and 1 atm pressure is $10^5\ \mathrm{Pa}$)

(A) 2.0 (B) 4.2 (C) 3.0 (D) 4.5

Open solution

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Q12. JEE Main 2026 (Online) 22nd January Evening Shift

When a part of a straight capillary tube is placed vertically in a liquid, the liquid raises upto certain height h. If the inner radius of the capillary tube, density of the liquid and surface tension of the liquid decrease by 1% each, then the height of the liquid in the tube will change by ____ $\%$.

Open solution

(A) +3 (B) +1 (C) -1 (D) -3

Q13. JEE Main 2026 (Online) 22nd January Morning Shift

Given below are two statements :

Statement I : Pressure of a fluid is exerted only on a solid surface in contact as the fluid-pressure does not exist everywhere in a still fluid.

Statement II : Excess potential energy of the molecules on the surface of a liquid, when compared to interior, results in surface tension.

In the light of the above statements, choose the correct answer from the options given below

Open solution

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q14. JEE Main 2026 (Online) 21st January Evening Shift

Surface tension of two liquids (having same densities), T_1 and T_2, are measured using capillary rise method utilizing two tubes with inner radii of r_1 and r_2 where $r_1 > r_2$. The measured liquid heights in these tubes are h_1 and h_2 respectively. [Ignore the weight of the liquid above the lowest point of miniscus]. The heights h_1 and h_2 and surface tensions T_1 and T_2 satisfy the relation :

Open solution

(A) $h_1 > h_2$ and $T_1 = T_2$ (B) $h_1 < h_2$ and $T_1 = T_2$

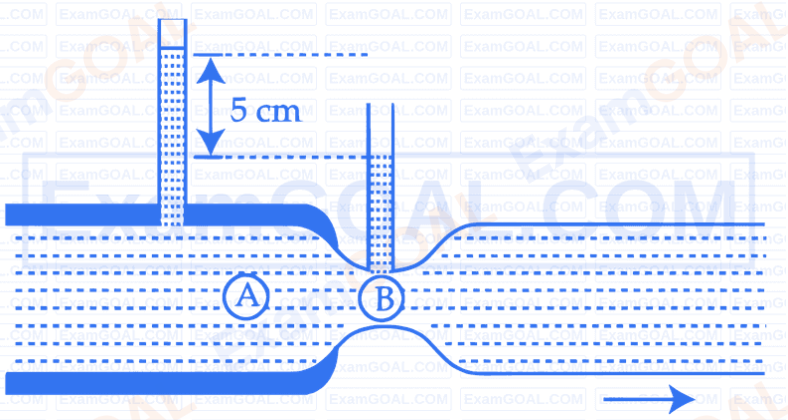
(C) $h_1 > h_2$ and $T_1 < T_2$ (D) $h_1 = h_2$ and $T_1 = T_2$

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Q15. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Water flows through a horizontal tube as shown in the figure. The difference in height between the water columns in vertical tubes is 5 cm and the area of cross-sections at A and B are $6\ \text{cm}^2$ and $3\ \text{cm}^2$ respectively. The rate of flow will be ____ cm^3/s. (take $g = 10\ \text{m/s}^2$)



(A) $100\sqrt{3}$ (B) $\frac{200}{\sqrt{3}}$ (C) $200\sqrt{6}$ (D) $200\sqrt{3}$

Q16. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

A capillary tube of radius 0.1 mm is partly dipped in water (surface tension 70 dyn/cm and glass water contact angle ≈ 0°) with 30° inclined with the vertical. The length of water risen in the capillary is _____ cm. (Take $g = 9.8\ \text{m/s}^2$)

(A) $\frac{82}{5}$ (B) $\frac{68}{5}$ (C) $\frac{57}{2}$ (D) $\frac{71}{5}$

Q17. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Two liquids A and B have θ_A and θ_B as contact angles in a capillary tube. If $K = \cos\theta_A / \cos\theta_B$, then identify the correct statement:

(A) K is negative, then liquid A and liquid B have convex meniscus.

(B) K is negative, then liquid A and liquid B have concave meniscus.

(C) K is zero, then liquid A has convex meniscus and liquid B has concave meniscus.

(D) K is negative, then liquid A has concave meniscus and liquid B has convex meniscus.

Q18. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

Two cylindrical vessels of equal cross sectional area of $2\ \text{m}^2$ contain water upto heights 10 m and 6 m , respectively. If the vessels are connected at their bottom then the work done by the force of gravity is (Density of water is $10^3\ \text{kg/m}^3$ and $\text{g} = 10\ \text{m/s}^2$)

(A) $1 \times 10^5\ \text{J}$ (B) $4 \times 10^4\ \text{J}$ (C) $8 \times 10^4\ \text{J}$ (D) $6 \times 10^4\ \text{J}$

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Q19. JEE Main 2025 (Online) 3rd April Evening Shift

A solid steel ball of diameter 3.6 mm acquired terminal velocity $2.45 \times 10^{-2}\ \mathrm{m/s}$ while falling under gravity through an oil of density $925\ \mathrm{kg\,m^{-3}}$. Take density of steel as $7825\ \mathrm{kg\,m^{-3}}$ and g as $9.8\ \mathrm{m/s^2}$. The viscosity of the oil in SI unit is

Open solution

(A) 2.18 (B) 1.68 (C) 2.38 (D) 1.99

Q20. JEE Main 2025 (Online) 3rd April Morning Shift

Consider a completely full cylindrical water tank of height 1.6 m and of cross-sectional area $0.5\ \mathrm{m^2}$. It has a small hole in its side at a height 90 cm from the bottom. Assume, the cross-sectional area of the hole to be negligibly small as compared to that of the water tank. If a load 50 kg is applied at the top surface of the water in the tank then the velocity of the water coming out at the instant when the hole is opened is:

$(\mathrm{g} = 10\ \mathrm{m/s^2})$

Open solution

(A) $2\ \mathrm{m/s}$ (B) $5\ \mathrm{m/s}$ (C) $3\ \mathrm{m/s}$ (D) $4\ \mathrm{m/s}$

Q21. JEE Main 2025 (Online) 2nd April Evening Shift

Two water drops each of radius ' r ' coalesce to form a bigger drop. If ' T ' is the surface tension, the surface energy released in this process is :

Open solution

(A) $4\pi \mathrm{r}^2\ \mathrm{T}\left[2 - 2^{1/3}\right]$ (B) $4\pi \mathrm{r}^2\ \mathrm{T}[1 + \sqrt{2}]$

(C) $4\pi \mathrm{r}^2\ \mathrm{T}\left[2 - 2^{2/3}\right]$ (D) $4\pi \mathrm{r}^2\ \mathrm{T}[\sqrt{2} - 1]$

Q22. JEE Main 2025 (Online) 28th January Evening Shift

A 400 g solid cube having an edge of length 10 cm floats in water. How much volume of the cube is outside the water?

(Given: density of water = 1000 kg m^{-3})

Open solution

(A) 400 cm^3 (B) 600 cm^3 (C) 1400 cm^3 (D) 4000 cm^3

Q23. JEE Main 2025 (Online) 28th January Morning Shift

In the experiment for measurement of viscosity ' η ' of given liquid with a ball having radius R, consider following statements.

Open solution

A. Graph between terminal velocity V and R will be a parabola.

B. The terminal velocities of different diameter balls are constant for a given liquid.

C. Measurement of terminal velocity is dependent on the temperature.

D. This experiment can be utilized to assess the density of a given liquid.

E. If balls are dropped with some initial speed, the value of η will change.

Choose the correct answer from the options given below:

(A) C, D and E Only (B) A, B and E Only (C) A, C and D Only (D) B, D and E Only

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Q24. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Consider following statements:

A. Surface tension arises due to extra energy of the molecules at the interior as compared to the molecules at the surface, of a liquid.

B. As the temperature of liquid rises, the coefficient of viscosity increases.

C. As the temperature of gas increases, the coefficient of viscosity increases

D. The onset of turbulence is determined by Reynold's number.

E. In a steady flow two stream lines never intersect.

Choose the correct answer from the options given below:

(A) B, C, D Only (B) C, D, E Only (C) A, D, E Only (D) A, B, C Only

Q25. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

An air bubble of radius 0.1 cm lies at a depth of 20 cm below the free surface of a liquid of density $1000\ \mathrm{kg/m^3}$. If the pressure inside the bubble is $2100\ \mathrm{N/m^2}$ greater than the atmospheric pressure, then the surface tension of the liquid in SI unit is (use $g = 10\ \mathrm{m/s^2}$)

(A) 0.02 (B) 0.05 (C) 0.25 (D) 0.1

Q26. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

The amount of work done to break a big water drop of radius ' R ' into 27 small drops of equal radius is 10 J . The work done required to break the same big drop into 64 small drops of equal radius will be

(A) 20 J (B) 10 J (C) 5 J (D) 15 J

Q27. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Water flows in a horizontal pipe whose one end is closed with a valve. The reading of the pressure gauge attached to the pipe is P_1. The reading of the pressure gauge falls to P_2 when the valve is opened. The speed of water flowing in the pipe is proportional to

(A) $(P_1 - P_2)^2$ (B) $\sqrt{\mathrm{P_1 - P_2}}$ (C) $P_1 - P_2$ (D) $(P_1 - P_2)^4$

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Q28. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Given below are two statements:

Statement I: The hot water flows faster than cold water

Statement II: Soap water has higher surface tension as compared to fresh water. In the light above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

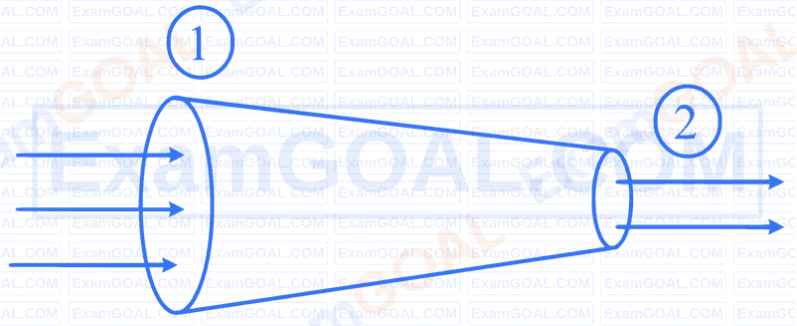
(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q29. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution



A tube of length L is shown in the figure. The radius of cross section at the point (1) is 2 cm and at the point (2) is 1 cm , respectively. If the velocity of water entering at point (1) is $2\ \mathrm{m/s}$, then velocity of water leaving the point (2) will be

(A) 4 m/s (B) 6 m/s (C) 2 m/s (D) 8 m/s

Q30. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

A small rigid spherical ball of mass M is dropped in a long vertical tube containing glycerine. The velocity of the ball becomes constant after some time. If the density of glycerine is half of the density of the ball, then the viscous force acting on the ball will be (consider g as acceleration due to gravity)

(A) 2 Mg (B) $\frac{3}{2}\mathrm{Mg}$ (C) $\frac{\mathrm{Mg}}{2}$ (D) Mg

Q31. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

The excess pressure inside a soap bubble is thrice the excess pressure inside a second soap bubble. The ratio between the volume of the first and the second bubble is:

(A) $1:9$ (B) $1:27$ (C) $1:81$ (D) $1:3$

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Q32. JEE Main 2024 (Online) 9th April Evening Shift

A spherical ball of radius $1 \times 10^{-4}\ \mathrm{m}$ and density $10^5\ \mathrm{kg/m^3}$ falls freely under gravity through a distance h before entering a tank of water. If after entering in water the velocity of the ball does not change, then the value of h is approximately:

(The coefficient of viscosity of water is $9.8 \times 10^{-6}\ \mathrm{N\,s/m^2}$)

(A) 2518 m (B) 2396 m (C) 2249 m (D) 2296 m

Open solution

Q33. JEE Main 2024 (Online) 9th April Morning Shift

A sphere of relative density σ and diameter D has concentric cavity of diameter d. The ratio of $\frac{D}{d}$, if it just floats on water in a tank is :

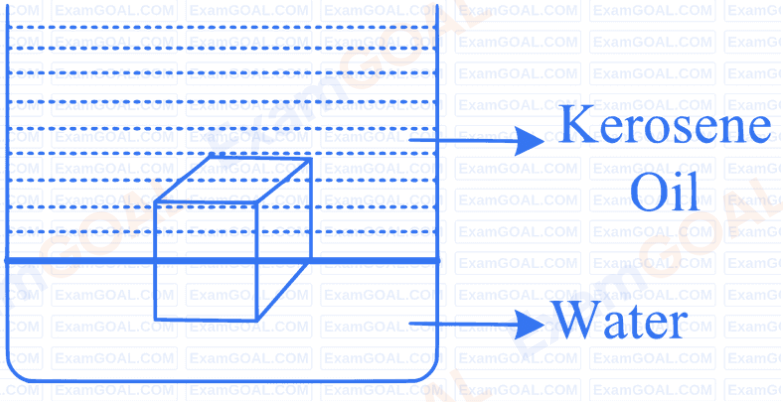
(A) $\left(\frac{\sigma-2}{\sigma+2}\right)^{1/3}$ (B) $\left(\frac{\sigma+1}{\sigma-1}\right)^{1/3}$

(C) $\left(\frac{\sigma-1}{\sigma}\right)^{1/3}$ (D) $\left(\frac{\sigma}{\sigma-1}\right)^{1/3}$

Open solution

Q34. JEE Main 2024 (Online) 8th April Evening Shift

A cube of ice floats partly in water and partly in kerosene oil. The ratio of volume of ice immersed in water to that in kerosene oil (specific gravity of Kerosene oil = 0.8, specific gravity of ice = 0.9):



(A) 5 : 4 (B) 9 : 10 (C) 8 : 9 (D) 1 : 1

Open solution

Q35. JEE Main 2024 (Online) 8th April Morning Shift

Correct Bernoulli's equation is (symbols have their usual meaning) :

(A) $P + \frac{1}{2}\rho gh + \frac{1}{2}\rho v^2 = \text{constant}$ (B) $P + mgh + \frac{1}{2}mv^2 = \text{constant}$

(C) $P + \rho gh + \rho v^2 = \text{constant}$ (D) $P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$

Open solution

Q36. JEE Main 2024 (Online) 6th April Evening Shift

Pressure inside a soap bubble is greater than the pressure outside by an amount : (given : R = Radius of bubble, S = Surface tension of bubble)

(A) $\frac{S}{R}$ (B) $\frac{4\,\mathrm{S}}{\mathrm{R}}$ (C) $\frac{4\mathrm{R}}{\mathrm{S}}$ (D) $\frac{2S}{R}$

Open solution

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Q37. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

A small ball of mass m and density ρ is dropped in a viscous liquid of density ρ_0. After sometime, the ball falls with constant velocity. The viscous force on the ball is :

(A) $mg\left(1-\frac{\rho_0}{\rho}\right)$

(B) $mg\left(\frac{\rho_0}{\rho}-1\right)$

(C) $mg(1-\rho\rho_0)$

(D) $mg\left(1+\frac{\rho}{\rho_0}\right)$

Q38. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

Given below are two statements :

Statement I : When a capillary tube is dipped into a liquid, the liquid neither rises nor falls in the capillary. The contact angle may be 0°.

Statement II : The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well.

In the light of the above statement, choose the correct answer from the options given below.

(A) Statement I is true and Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q39. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

Given below are two statements :

Statement I : The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well.

Statement II : The rise of a liquid in a capillary tube does not depend on the inner radius of the tube.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false.

(B) Both Statement I and Statement II are true.

(C) Statement I is false but Statement II is true.

(D) Statement I is true but Statement II is false.

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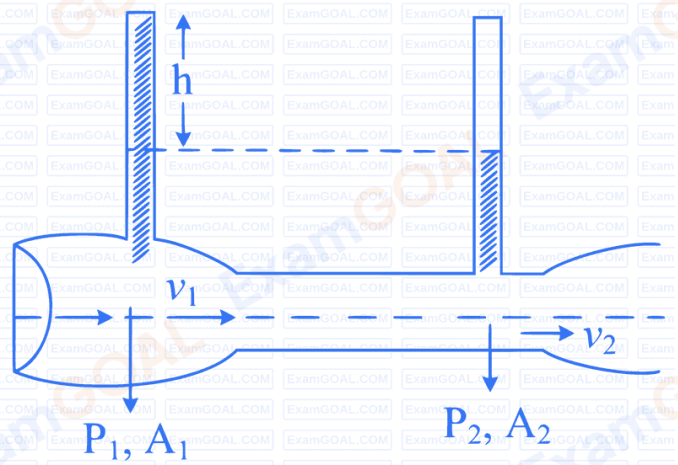
Q40. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

Given below are two statements :

Statement I : When speed of liquid is zero everywhere, pressure difference at any two points depends on equation $\mathrm{P}_1 - \mathrm{P}_2 = \rho g\,(\,\mathrm{h}_2 - \mathrm{h}_1)$.

Statement II : In ventury tube shown $2\mathrm{gh} = v_1^2 - v_2^2$



In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Statement I is correct but Statement II is incorrect.

(B) Both Statement I and Statement II are correct.

(C) Both Statement I and Statement II are incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q41. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

A big drop is formed by coalescing 1000 small droplets of water. The surface energy will become :

(A) $\frac{1}{100}$ th (B) $\frac{1}{10}$ th (C) 100 times (D) 10 times

Q42. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

A small spherical ball of radius r, falling through a viscous medium of negligible density has terminal velocity 'v'. Another ball of the same mass but of radius $2r$, falling through the same viscous medium will have terminal velocity:

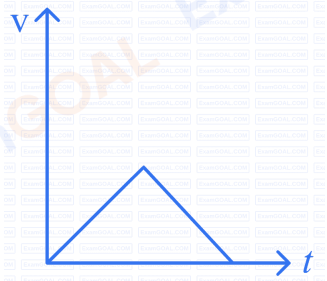
(A) $4\mathrm{v}$ (B) $2\mathrm{v}$ (C) $\frac{v}{4}$ (D) $\frac{\mathrm{v}}{2}$

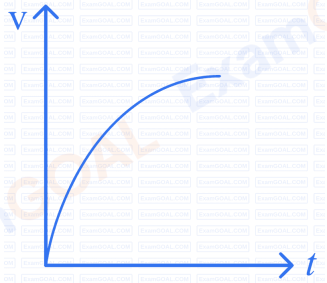
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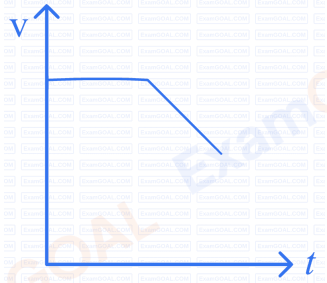
Q43. JEE Main 2024 (Online) 31st January Morning Shift

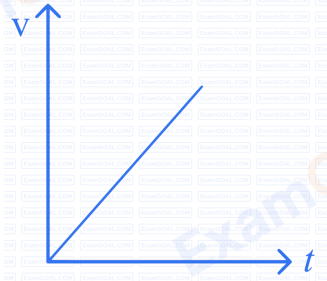
A small steel ball is dropped into a long cylinder containing glycerine. Which one of the following is the correct representation of the velocity time graph for the transit of the ball?

Open solution

(A) 

(B) 

(C) 

(D) 

Q44. JEE Main 2024 (Online) 29th January Evening Shift

A small liquid drop of radius R is divided into 27 identical liquid drops. If the surface tension is T, then the work done in the process will be:

(A) $4\pi \mathrm{R}^2\ \mathrm{T}$ (B) $8\pi R^2\ \mathrm{T}$ (C) $\frac{1}{8}\pi R^2 T$ (D) $3\pi R^2\ \mathrm{T}$

Open solution

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Q45. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Given below are two statements:

Statement I : If a capillary tube is immersed first in cold water and then in hot water, the height of capillary rise will be smaller in hot water.

Statement II : If a capillary tube is immersed first in cold water and then in hot water, the height of capillary rise will be smaller in cold water.

In the light of the above statements, choose the most appropriate from the options given below

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q46. JEE Main 2024 (Online) 27th January Morning Shift

Open solution

Given below are two statements :

Statement (I) :Viscosity of gases is greater than that of liquids.

Statement (II) : Surface tension of a liquid decreases due to the presence of insoluble impurities.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are incorrect

(D) Both Statement I and Statement II are correct

Q47. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : A spherical body of radius $(5 \pm 0.1)\mathrm{mm}$ having a particular density is falling through a liquid of constant density. The percentage error in the calculation of its terminal velocity is 4%.

Reason R : The terminal velocity of the spherical body falling through the liquid is inversely proportional to its radius.

In the light of the above statements, choose the correct answer from the options given below

(A) A is false but $\mathbf{R}$ is true

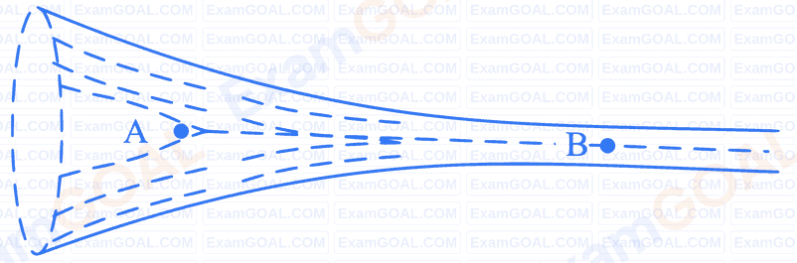
(B) A is true but $\mathbf{R}$ is false

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

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Q48. JEE Main 2023 (Online) 13th April Morning Shift



Open solution

The figure shows a liquid of given density flowing steadily in horizontal tube of varying cross - section. Cross sectional areas at A is $1.5\ \mathrm{cm}^2$, and B is $25\ \mathrm{mm}^2$, if the speed of liquid at B is $60\ \mathrm{cm/s}$ then $(\mathrm{P_A} - \mathrm{P_B})$ is :

(Given $\mathrm{P_A}$ and $\mathrm{P_B}$ are liquid pressures at A and B points.

density $\rho = 1000\ \mathrm{kg\,m^{-3}}$

A and B are on the axis of tube

(A) $27\ \mathrm{Pa}$ (B) $175\ \mathrm{Pa}$ (C) $135\ \mathrm{Pa}$ (D) $36\ \mathrm{Pa}$

Q49. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

Eight equal drops of water are falling through air with a steady speed of $10\ \mathrm{cm/s}$. If the drops coalesce, the new velocity is:-

(A) $40\ \mathrm{cm/s}$ (B) $16\ \mathrm{cm/s}$ (C) $10\ \mathrm{cm/s}$ (D) $5\ \mathrm{cm/s}$

Q50. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

Given below are two statements:

Statement I : Pressure in a reservoir of water is same at all points at the same level of water.

Statement II : The pressure applied to enclosed water is transmitted in all directions equally.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

Q51. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

A hydraulic automobile lift is designed to lift vehicles of mass $5000\ \mathrm{kg}$. The area of cross section of the cylinder carrying the load is $250\ \mathrm{cm}^2$. The maximum pressure the smaller piston would have to bear is $[$ Assume $g = 10\ \mathrm{m/s^2}]$

(A) $20 \times 10^{+6}\ \mathrm{Pa}$ (B) $200 \times 10^{+6}\ \mathrm{Pa}$ (C) $2 \times 10^{+5}\ \mathrm{Pa}$ (D) $2 \times 10^{+6}\ \mathrm{Pa}$

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Q52. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

An air bubble of volume $1\ \text{cm}^3$ rises from the bottom of a lake $40\ \text{m}$ deep to the surface at a temperature of 12°C. The atmospheric pressure is $1 \times 10^5\ \text{Pa}$ the density of water is $1000\ \text{kg/m}^3$ and $g = 10\ \text{m/s}^2$. There is no difference of the temperature of water at the depth of $40\ \text{m}$ and on the surface. The volume of air bubble when it reaches the surface will be:

(A) $4\ \text{cm}^3$ (B) $3\ \text{cm}^3$ (C) $2\ \text{cm}^3$ (D) $5\ \text{cm}^3$

Q53. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A: When you squeeze one end of a tube to get toothpaste out from the other end, Pascal's principle is observed.

Reason R: A change in the pressure applied to an enclosed incompressible fluid is transmitted undiminished to every portion of the fluid and to the walls of its container.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Both A and R are correct but R is NOT the correct explanation of A

(B) A is not correct but R is correct

(C) A is correct but R is not correct

(D) Both A and B are correct and R is the correct explanation of A

Q54. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

A small ball of mass M and density ρ is dropped in a viscous liquid of density ρ_0. After some time, the ball falls with a constant velocity. What is the viscous force on the ball ?

(A) $\text{F} = \text{Mg}\left(1 - \frac{\rho_0}{\rho}\right)$ (B) $\text{F} = \text{Mg}\left(1 + \frac{\rho}{P_o}\right)$

(C) $\text{F} = \text{Mg}\left(1 + \frac{\rho_o}{\rho}\right)$ (D) $F = Mg\,(1 \pm \rho\rho_0)$

Q55. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

A mercury drop of radius $10^{-3}\ \text{m}$ is broken into 125 equal size droplets. Surface tension of mercury is $0.45\ \text{Nm}^{-1}$. The gain in surface energy is :

(A) $28 \times 10^{-5}\ \text{J}$ (B) $17.5 \times 10^{-5}\ \text{J}$ (C) $5 \times 10^{-5}\ \text{J}$ (D) $2.26 \times 10^{-5}\ \text{J}$

Q56. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

If 1000 droplets of water of surface tension $0.07\ \text{N/m}$, having same radius $1\ \text{mm}$ each, combine to form a single drop. In the process the released surface energy is :-

$\left(\text{Take}\ \pi = \frac{22}{7}\right)$

(A) $7.92 \times 10^{-4}\ \text{J}$ (B) $7.92 \times 10^{-6}\ \text{J}$ (C) $8.8 \times 10^{-5}\ \text{J}$ (D) $9.68 \times 10^{-4}\ \text{J}$

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Q57. JEE Main 2023 (Online) 30th January Morning Shift

The height of liquid column raised in a capillary tube of certain radius when dipped in liquid A vertically is, $5\ \mathrm{cm}$. If the tube is dipped in a similar manner in another liquid B of surface tension and density double the values of liquid A, the height of liquid column raised in liquid B would be _____ m.

Open solution

(A) 0.05 (B) 0.20 (C) 0.5 (D) 0.10

Q58. JEE Main 2023 (Online) 29th January Evening Shift

A fully loaded boeing aircraft has a mass of 5.4×10^5 kg. Its total wing area is 500 m^2. It is in level flight with a speed of 1080 km/h. If the density of air ρ is 1.2 kg m^{-3}, the fractional increase in the speed of the air on the upper surface of the wing relative to the lower surface in percentage will be. $(\mathrm{g} = 10\ \mathrm{m/s^2})$

Open solution

(A) 16 (B) 8 (C) 6 (D) 10

Q59. JEE Main 2023 (Online) 29th January Morning Shift

Surface tension of a soap bubble is $2.0 \times 10^{-2}\mathrm{Nm}^{-1}$. Work done to increase the radius of soap bubble from $3.5\ \mathrm{cm}$ to $7\ \mathrm{cm}$ will be:

Take $\left[\pi = \frac{22}{7}\right]$

Open solution

(A) $18.48 \times 10^{-4}\ \mathrm{J}$ (B) $5.76 \times 10^{-4}\ \mathrm{J}$ (C) $0.72 \times 10^{-4}\ \mathrm{J}$ (D) $9.24 \times 10^{-4}\ \mathrm{J}$

Q60. JEE Main 2022 (Online) 29th July Morning Shift

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Clothes containing oil or grease stains cannot be cleaned by water wash.

Reason (R): Because the angle of contact between the oil/grease and water is obtuse.

In the light of the above statements, choose the correct answer from the option given below.

Open solution

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Q61. JEE Main 2022 (Online) 28th July Evening Shift

A pressure-pump has a horizontal tube of cross sectional area $10\ \mathrm{cm}^2$ for the outflow of water at a speed of $20\ \mathrm{m/s}$. The force exerted on the vertical wall just in front of the tube which stops water horizontally flowing out of the tube, is :

[given: density of water $= 1000\ \mathrm{kg/m^3}$]

Open solution

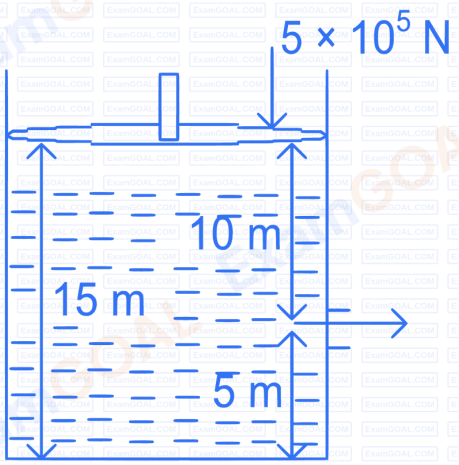
(A) 300 N (B) 500 N (C) 250 N (D) 400 N

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Q62. JEE Main 2022 (Online) 28th July Evening Shift

Open solution

Consider a cylindrical tank of radius $1\ \mathrm{m}$ is filled with water. The top surface of water is at $15\ \mathrm{m}$ from the bottom of the cylinder. There is a hole on the wall of cylinder at a height of $5\ \mathrm{m}$ from the bottom. A force of $5 \times 10^5\ \mathrm{N}$ is applied on the top surface of water using a piston. The speed of efflux from the hole will be : (given atmospheric pressure $\mathrm{P_A} = 1.01 \times 10^5\ \mathrm{Pa}$, density of water $\rho_{\mathrm{W}} = 1000\ \mathrm{kg/m^3}$ and gravitational acceleration $\mathrm{g} = 10\ \mathrm{m/s^2}$)



(A) 11.6 m/s (B) 10.8 m/s (C) 17.8 m/s (D) 14.4 m/s

Q63. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

Two cylindrical vessels of equal cross-sectional area $16\ \mathrm{cm^2}$ contain water upto heights $100\ \mathrm{cm}$ and $150\ \mathrm{cm}$ respectively. The vessels are interconnected so that the water levels in them become equal. The work done by the force of gravity during the process, is [Take, density of water $= 10^3\ \mathrm{kg/m^3}$ and $\mathrm{g} = 10\ \mathrm{ms^{-2}}$] :

(A) 0.25 J (B) 1 J (C) 8 J (D) 12 J

Q64. JEE Main 2022 (Online) 26th July Morning Shift

Open solution

A water drop of radius $1\ \mathrm{cm}$ is broken into 729 equal droplets. If surface tension of water is 75 dyne/ cm, then the gain in surface energy upto first decimal place will be :

(Given $\pi = 3.14$)

(A) $8.5 \times 10^{-4}\ \mathrm{J}$ (B) $8.2 \times 10^{-4}\ \mathrm{J}$ (C) $7.5 \times 10^{-4}\ \mathrm{J}$ (D) $5.3 \times 10^{-4}\ \mathrm{J}$

Q65. JEE Main 2022 (Online) 25th July Evening Shift

Open solution

A drop of liquid of density ρ is floating half immersed in a liquid of density σ and surface tension 7.5×10^{-4} Ncm^{-1}. The radius of drop in cm will be :

(g = 10 ms^{-2})

(A) $\frac{15}{\sqrt{(2\rho-\sigma)}}$ (B) $\frac{15}{\sqrt{(\rho-\sigma)}}$

(C) $\frac{3}{2\sqrt{(\rho-\sigma)}}$ (D) $\frac{3}{20\sqrt{(2\rho-\sigma)}}$

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Q66. JEE Main 2022 (Online) 30th June Morning Shift

An air bubble of negligible weight having radius r rises steadily through a solution of density σ at speed v. The coefficient of viscosity of the solution is given by :

(A) $\eta = \frac{4r\sigma g}{9v}$

(B) $\eta = \frac{2r^2\sigma g}{9v}$

(C) $\eta = \frac{2\pi r^2\sigma g}{9v}$

(D) $\eta = \frac{2r^2\sigma g}{3\pi v}$

Open solution

Q67. JEE Main 2022 (Online) 28th June Evening Shift

A water drop of radius 1 μm falls in a situation where the effect of buoyant force is negligible. Co-efficient of viscosity of air is 1.8×10^{-5} Nsm^{-2} and its density is negligible as compared to that of water 10^6 gm^{-3}. Terminal velocity of the water drop is :

(Take acceleration due to gravity = 10 ms^{-2})

(A) 145.4×10^{-6} ms^{-1}

(B) 118.0×10^{-6} ms^{-1}

(C) 132.6×10^{-6} ms^{-1}

(D) 123.4×10^{-6} ms^{-1}

Open solution

Q68. JEE Main 2022 (Online) 28th June Morning Shift

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Product of Pressure (P) and time (t) has the same dimension as that of coefficient of viscosity.

Reason R : Coefficient of viscosity = $\frac{Force}{Velocity\ gradient}$

Choose the correct answer from the options given below :

(A) Both A and R are true, and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Open solution

Q69. JEE Main 2022 (Online) 28th June Morning Shift

A water drop of diameter 2 cm is broken into 64 equal droplets. The surface tension of water is 0.075 N/m. In this process the gain in surface energy will be :

(A) 2.8×10^{-4} J

(B) 1.5×10^{-3} J

(C) 1.9×10^{-4} J

(D) 9.4×10^{-5} J

Open solution

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Q70. JEE Main 2022 (Online) 27th June Morning Shift

The velocity of a small ball of mass 'm' and density d_1, when dropped in a container filled with glycerin, becomes constant after some time. If the density of glycerin is d_2, then the viscous force acting on the ball, will be :

(A) $mg\left(1-\frac{d_1}{d_2}\right)$ (B) $mg\left(1-\frac{d_2}{d_1}\right)$

(C) $mg\left(\frac{d_1}{d_2}-1\right)$ (D) $mg\left(\frac{d_2}{d_1}-1\right)$

Open solution

Q71. JEE Main 2022 (Online) 26th June Evening Shift

If ρ is the density and η is coefficient of viscosity of fluid which flows with a speed v in the pipe of diameter d, the correct formula for Reynolds number R_e is :

(A) $R_e=\frac{\eta d}{\rho v}$ (B) $R_e=\frac{\rho v}{\eta d}$

(C) $R_e=\frac{\rho v d}{\eta}$ (D) $R_e=\frac{\eta}{\rho v d}$

Open solution

Q72. JEE Main 2022 (Online) 25th June Morning Shift

The terminal velocity (v_t) of the spherical rain drop depends on the radius (r) of the spherical rain drop as :

(A) $r^{1/2}$ (B) r (C) r^2 (D) r^3

Open solution

Q73. JEE Main 2021 (Online) 27th August Morning Shift

In Millikan's oil drop experiment, what is viscous force acting on an uncharged drop of radius 2.0×10^{-5} m and density 1.2×10^3 kgm^{-3} ? Take viscosity of liquid = 1.8×10^{-5} Nsm^{-2}. (Neglect buoyancy due to air).

(A) 3.8×10^{-11} N (B) 3.9×10^{-10} N (C) 1.8×10^{-10} N (D) 5.8×10^{-10} N

Open solution

Q74. JEE Main 2021 (Online) 26th August Morning Shift

Two narrow bores of diameter 5.0 mm and 8.0 mm are joined together to form a U-shaped tube open at both ends. If this U-tube contains water, what is the difference in the level of two limbs of the tube. [Take surface tension of water T = 7.3×10^{-2} Nm^{-1}, angle of contact = 0, g = 10 ms^{-2} and density of water = 1.0×10^3 kg m^{-3}]

(A) 3.62 mm (B) 2.19 mm (C) 5.34 mm (D) 4.97 mm

Open solution

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Q75. JEE Main 2021 (Online) 27th July Evening Shift

A raindrop with radius R = 0.2 mm falls from a cloud at a height h = 2000 m above the ground. Assume that the drop is spherical throughout its fall and the force of buoyance may be neglected, then the terminal speed attained by the raindrop is :

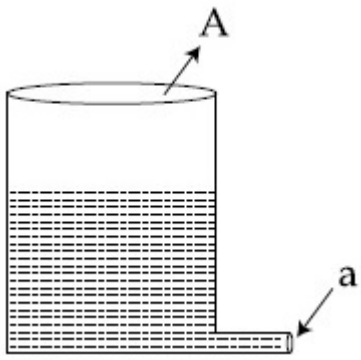
[Density of water f_w = 1000 kg m^{-3} and Density of air f_a = 1.2 kg m^{-3}, g = 10 m/s^2, Coefficient of viscosity of air = 1.8 $\times$ 10^{-5} Nsm^{-2}]

(A) 250.6 ms^{-1} (B) 43.56 ms^{-1} (C) 4.94 ms^{-1} (D) 14.4 ms^{-1}

Open solution

Q76. JEE Main 2021 (Online) 27th July Morning Shift

A light cylindrical vessel is kept on a horizontal surface. Area of base is A. A hole of cross-sectional area 'a' is made just at its bottom side. The minimum coefficient of friction necessary to prevent sliding the vessel due to the impact force of the emerging liquid is (a < < A) :



(A) $\frac{A}{2a}$ (B) None of these (C) $\frac{2a}{A}$ (D) $\frac{a}{A}$

Open solution

Q77. JEE Main 2021 (Online) 25th July Evening Shift

Two spherical soap bubbles of radii r_1 and r_2 in vacuum combine under isothermal conditions. The resulting bubble has a radius equal to :

(A) $\frac{r_1 r_2}{r_1+r_2}$ (B) $\sqrt{r_1 r_2}$

(C) $\sqrt{r_1^2+r_2^2}$ (D) $\frac{r_1+r_2}{2}$

Open solution

Q78. JEE Main 2021 (Online) 20th July Evening Shift

Two small drops of mercury each of radius R coalesce to form a single large drop. The ratio of total surface energy before and after the change is :

(A) $2^{\frac{1}{3}} : 1$ (B) $1 : 2^{\frac{1}{3}}$ (C) 2 : 1 (D) 1 : 2

Open solution

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Q79. JEE Main 2021 (Online) 17th March Morning Shift

When two soap bubbles of radii a and b (b > a) coalesce, the radius of curvature of common surface is :

(A) $\frac{b-a}{ab}$ (B) $\frac{a+b}{ab}$ (C) $\frac{ab}{a+b}$ (D) $\frac{ab}{b-a}$

Open solution

Q80. JEE Main 2021 (Online) 16th March Evening Shift

What will be the nature of flow of water from a circular tap, when its flow rate increased from 0.18 L/min to 0.48 L/min? The radius of the tap and viscosity of water are 0.5 cm and 10^{-3} Pa s, respectively. (Density of water : 10^3 kg/m^3)

(A) Steady flow to unsteady flow
(B) Unsteady to steady flow
(C) Remains turbulent flow
(D) Remains steady flow

Open solution

Q81. JEE Main 2021 (Online) 16th March Morning Shift

The pressure acting on a submarine is 3×10^5 Pa at a certain depth. If the depth is doubled, the percentage increase in the pressure acting on the submarine would be :

(Assume that atmospheric pressure is 1×10^5 Pa and density of water is 10^3 kg m^{-3}, g = 10 ms^{-2})

(A) $\frac{200}{5}$ % (B) $\frac{200}{3}$ % (C) $\frac{3}{200}$ % (D) $\frac{5}{200}$ %

Open solution

Q82. JEE Main 2021 (Online) 26th February Morning Shift

A large number of water drops, each of radius r, combine to have a drop of radius R. If the surface tension is T and mechanical equivalent of heat is J, the rise in heat energy per unit volume will be :

(A) $\frac{2T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$ (B) $\frac{3T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$
(C) $\frac{3T}{rJ}$ (D) $\frac{2T}{rJ}$

Open solution

Q83. JEE Main 2020 (Online) 6th September Evening Slot

A fluid is flowing through a horizontal pipe of varying cross-section, with speed v ms^{-1} at a point where the pressure is P pascal. At another point where pressure is $\frac{P}{2}$ Pascal its speed is V ms^{-1}. If the density of the fluid is ρ kg m^{-3} and the flow is streamline, then V is equal to :

(A) $\sqrt{\frac{P}{2\rho}+v^2}$ (B) $\sqrt{\frac{P}{\rho}+v^2}$
(C) $\sqrt{\frac{2P}{\rho}+v^2}$ (D) $\sqrt{\frac{P}{\rho}+v}$

Open solution

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Q84. JEE Main 2020 (Online) 5th September Evening Slot

In an experiment to verify Stokes law, a small spherical ball of radius r and density ρ falls under gravity through a distance h in air before entering a tank of water. If the terminal velocity of the ball inside water is same as its velocity just before entering the water surface, then the value of h is proportional to :
(ignore viscosity of air)

(A) r (B) r^4 (C) r^3 (D) r^2

Open solution

Q85. JEE Main 2020 (Online) 5th September Morning Slot

A hollow spherical shell of outer radius R floats just submerged under the water surface. The inner radius of the shell is r. If the specific gravity of the shell material is $\frac{27}{8}$ w.r.t water, the value of r is :

(A) $\frac{2}{3}$R (B) $\frac{4}{9}$R (C) $\frac{1}{3}$R (D) $\frac{8}{9}$R

Open solution

Q86. JEE Main 2020 (Online) 4th September Evening Slot

Two identical cylindrical vessels are kept on the ground and each contain the same liquid of density d. The area of the base of both vessels is S but the height of liquid in one vessel is x_1 and in the other, x_2 . When both cylinders are connected through a pipe of negligible volume very close to the bottom, the liquid flows from one vessel to the other until it comes to equilibrium at a new height. The change in energy of the system in the process is:

(A) $gdS(x_2 + x_1)^2$ (B) $gdS\left(x_2^2 + x_1^2\right)$

(C) $\frac{1}{4}gdS(x_2 - x_1)^2$ (D) $\frac{3}{4}gdS(x_2 - x_1)^2$

Open solution

Q87. JEE Main 2020 (Online) 4th September Morning Slot

A air bubble of radius 1 cm in water has an upward acceleration 9.8 cm s^{-2}. The density of water is 1 gm cm^{-3} and water offers negligible drag force on the bubble. The mass of the bubble is (g = 980 cm/s^2).

(A) 1.52 gm (B) 4.51 gm (C) 3.15 gm (D) 4.15 gm

Open solution

Q88. JEE Main 2020 (Online) 3rd September Morning Slot

Pressure inside two soap bubbles are 1.01 and 1.02 atmosphere, respectively. The ratio of their volumes is :

(A) 4 : 1 (B) 8 : 1 (C) 2 : 1 (D) 0.8 : 1

Open solution

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Q89. JEE Main 2020 (Online) 2nd September Evening Slot

A capillary tube made of glass of radius 0.15 mm is dipped vertically in a beaker filled with methylene iodide (surface tension = 0.05 Nm^{-1}, density = 667 kg m^{-3}) which rises to height h in the tube. It is observed that the two tangents drawn from liquid-glass interfaces (from opp. sides of the capillary) make an angle of 60° with one another. Then h is close to (g = 10 ms^{-2})

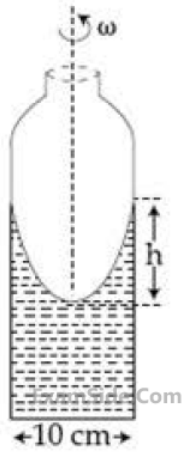
Open solution

(A) 0.049 m (B) 0.087 m (C) 0.137 m (D) 0.172 m

Q90. JEE Main 2020 (Online) 2nd September Morning Slot

A cylindrical vessel containing a liquid is rotated about its axis so that the liquid rises at its sides as shown in the figure. The radius of vessel is 5 cm and the angular speed of rotation is ω rad s^{-1}. The difference in the height, h (in cm) of liquid at the centre of vessel and at the side will be :

Open solution



(A) $\frac{2\omega^2}{25g}$ (B) $\frac{5\omega^2}{2g}$ (C) $\frac{25\omega^2}{2g}$ (D) $\frac{2\omega^2}{5g}$

Q91. JEE Main 2020 (Online) 9th January Evening Slot

A small spherical droplet of density d is floating exactly half immersed in a liquid of density ρ and surface tension T. The radius of the droplet is (take note that the surface tension applies an upward force on the droplet) :

Open solution

(A) $r = \sqrt{\frac{T}{(d-\rho)g}}$ (B) $r = \sqrt{\frac{2T}{3(d+\rho)g}}$

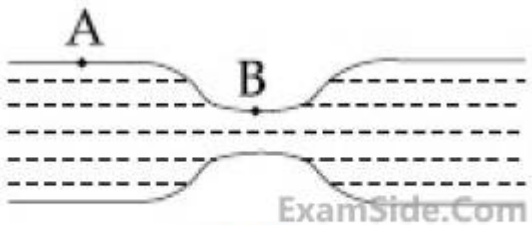
(C) $r = \sqrt{\frac{T}{(d+\rho)g}}$ (D) $r = \sqrt{\frac{3T}{(2d-\rho)g}}$

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Q92. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

Water flows in a horizontal tube (see figure). The pressure of water changes by 700 Nm^{-2} between A and B where the area of cross section are 40 cm^2 and 20 cm^2, respectively. Find the rate of flow of water through the tube. (density of water = 1000 kgm^{-3})



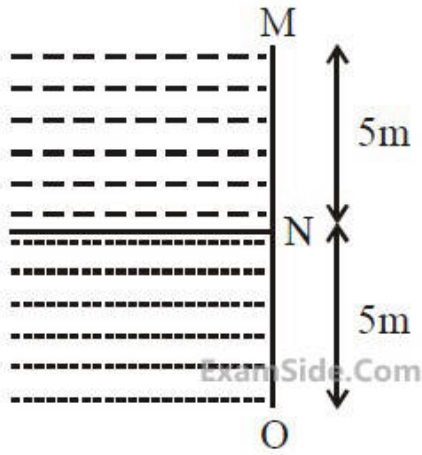
(Fig)

(A) 1810 cm^3/s (B) 2420 cm^3/s (C) 3020 cm^3/s (D) 2720 cm^3/s

Q93. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

Two liquids of densities ρ_1 and ρ_2 ($\rho_2 = 2\rho_1$) are filled up behind a square wall of side 10 m as shown in figure. Each liquid has a height of 5 m. The ratio of the forces due to these liquids exerted on upper part MN to that at the lower part NO is (Assume that the liquids are not mixing)



(A) 1/3 (B) 1/2 (C) 1/4 (D) 2/3

Q94. JEE Main 2020 (Online) 8th January Morning Slot

Open solution

Consider a solid sphere of radius R and mass density $\rho(r) = \rho_0\left(1 - \frac{r^2}{R^2}\right), 0 < r \leq R$

The minimum density of a liquid in which it will float is :

(A) $\frac{2\rho_0}{3}$ (B) $\frac{2\rho_0}{5}$ (C) $\frac{\rho_0}{5}$ (D) $\frac{\rho_0}{3}$

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Q95. JEE Main 2020 (Online) 8th January Morning Slot

A leak proof cylinder of length 1m, made of a metal which has very low coefficient of expansion is floating vertically in water at 0°C such that its height above the water surface is 20 cm. When the temperature of water is increased to 4°C, the height of the cylinder above the water surface becomes 21 cm. The density of water at T = 4°C, relative to the density at T = 0°C is close to :

(A) 1.04 (B) 1.26 (C) 1.01 (D) 1.03

Open solution

Q96. JEE Main 2020 (Online) 7th January Evening Slot

An ideal fluid flows (laminar flow) through a pipe of non-uniform diameter. The maximum and minimum diameters of the pipes are 6.4 cm and 4.8 cm, respectively. The ratio of the minimum and the maximum velocities of fluid in this pipe is :

(A) $\frac{3}{4}$ (B) $\frac{9}{16}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{81}{256}$

Open solution

Q97. JEE Main 2019 (Online) 12th April Evening Slot

A solid sphere, of radius R acquires a terminal velocity v_1 when falling (due to gravity) through a viscous fluid having a coefficient of viscosity . The sphere is broken into 27 identical solid spheres. If each of these spheres acquires a terminal velocity, v_2, when falling through the same fluid, the ratio (v_1/v_2) equals :

(A) $\frac{1}{9}$ (B) $\frac{1}{27}$ (C) 27 (D) 9

Open solution

Q98. JEE Main 2019 (Online) 10th April Evening Slot

Water from a tap emerges vertically downwards with an initial speed of 1.0 ms^{-1} . The cross-sectional area of the tap is 10^{-4} m^2. Assume that the pressure is constant throughout the stream of water and that the flow is streamlined. The cross-sectional area of the stream, 0.15 m below the tap would be : (Take g = 10 ms^{-2})

(A) 5×10^{-4} m^2 (B) 2×10^{-5} m^2 (C) 5×10^{-5} m^2 (D) 1×10^{-5} m^2

Open solution

Q99. JEE Main 2019 (Online) 10th April Evening Slot

A submarine experiences a pressure of 5.05×10^6 Pa at a depth of d_1 in a sea. When it goes further to a depth of d_2, it experiences a pressure of 8.08×10^6 Pa. Then $d_2 - d_1$ is approximately (density of water = 10^3 kg/m^3 and acceleration due to gravity = 10 ms^{-2}) :

(A) 600 m (B) 400 m (C) 300 m (D) 500 m

Open solution

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Q100. JEE Main 2019 (Online) 10th April Evening Slot

A cubical block of side 0.5 m floats on water with 30% of its volume under water. What is the maximum weight that can be put on the block without fully submerging it under water? [Take, density of water = 10^3 kg/m^3]

Open solution

(A) 30.1 kg (B) 87.5 kg (C) 65.4 kg (D) 46.3 kg

Q101. JEE Main 2019 (Online) 10th April Morning Slot

The ratio of surface tensions of mercury and water is given to be 7.5 while the ratio of their densities is 13.6. Their contact angles, with glass, are close to 135° and 0°, respectively. It is observed that mercury gets depressed by an amount h in a capillary tube of radius r_1, while water rises by the same amount h in a capillary tube of radius r_2. The ratio, (r_1/r_2), is then close to :

Open solution

(A) 2/5 (B) 2/3 (C) 3/5 (D) 4/5

Q102. JEE Main 2019 (Online) 9th April Evening Slot

A wooden block floating in a bucket of water has 4/5 of its volume submerged. When certain amount of an oil is poured into the bucket, it is found that the block is just under the oil surface with half of its volume under water and half in oil. The density of oil relative to that of water is :-

Open solution

(A) 0.8 (B) 0.7 (C) 0.6 (D) 0.5

Q103. JEE Main 2019 (Online) 9th April Morning Slot

If 'M' is the mass of water that rises in a capillary tube of radius 'r', then mass of water which will rise in a capillary tube of radius '2r' is :

Open solution

(A) M (B) 4M (C) M/2 (D) 2M

Q104. JEE Main 2019 (Online) 8th April Morning Slot

Water from a pipe is coming at a rate of 100 litres per minute. If the radius of the pipe is 5 cm, the Reynolds number for the flow is of the order of : (density of water = 1000 kg/m^3, coefficient of viscosity of water = 1mPas)

Open solution

(A) 10^6 (B) 10^4 (C) 10^3 (D) 10^2

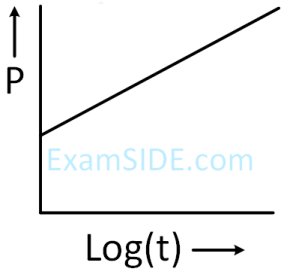
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Q105. JEE Main 2019 (Online) 12th January Evening Slot

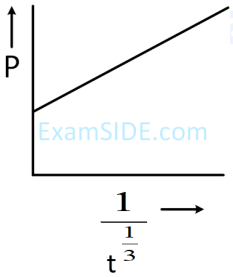
A soap bubble, blown by a mechanical pump at the mouth of a tube, increases in volume, with time, at a constant rate. The graph that correctly depicts the time dependence of pressure inside the bubble is given by :

Open solution

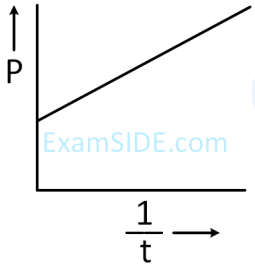
(A)



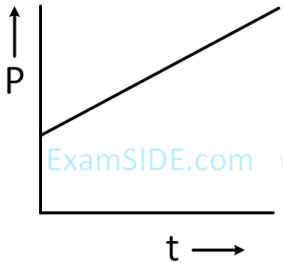
(B)



(C)



(D)



Q106. JEE Main 2019 (Online) 12th January Evening Slot

A long cylindrical vessel is half filled with a liquid. When the vessel is rotated about its own vertical axis, the liquid rises up near the wall. If the radius of vessel is 5 cm and its rotational speed is 2 rotations per second, then the difference in the heights between the centre and the sides, in cm, will be :

Open solution

(A) 2.0 (B) 1.2 (C) 0.1 (D) 0.4

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Q107. JEE Main 2019 (Online) 11th January Morning Slot

A liquid of density ρ is coming out of a hose pipe of radius a with horizontal speed v and hits a mesh. 50% of the liquid passes through the mesh unaffected. 25% looses all of its momentum and 25% comes back with the same speed. The resultant pressure on the mesh will be :

Open solution

(A) $\frac{3}{4}\rho v^2$ (B) $\frac{1}{4}\rho v^2$ (C) $\frac{1}{2}\rho v^2$ (D) ρv^2

Q108. JEE Main 2019 (Online) 10th January Morning Slot

Water flows into a large tank with flat bottom at the rate of $10^{-4}m^3s^{-1}$. Water is also leaking out of a hole ofarea 1 cm^2 at its bottom. If the height of the water in the tank remains steady, then this height is -

Open solution

(A) 2.9 cm (B) 5.1 cm (C) 4 cm (D) 1.7 cm

Q109. JEE Main 2019 (Online) 9th January Evening Slot

The top of a water tank is open to air and its water level is mainted. It is giving out 0.74 m^3 water per minute through a circular opening of 2 cm radius in its wall. The depth of the center of the opening from the level of water in the tank is close to :

Open solution

(A) 6.0 m (B) 4.8 m (C) 9.6 m (D) 2.9 m

Q110. JEE Main 2018 (Online) 16th April Morning Slot

A small soap bubble of radius 4 cm is trapped inside another bubble of radius 6 cm without any contact. Let P_2 be the pressure inside the inner bubble and P_0, the pressure outside the outer bubble. Radius of another bubble with pressure difference $P_2 - P_0$ between its inside and outside would be :

Open solution

(A) 12 cm (B) 2.4 cm (C) 6 cm (D) 4.8 cm

Q111. JEE Main 2018 (Online) 15th April Evening Slot

When an air bubble of radius r rises from the bottom to the surface of a lake its radius becomes $\frac{5r}{4}$. Taking the atmospheric pressure to be equal to 10 m height of water column, the depth of the lake would approximately be (ignore the surface tension and the effect of temperature) :

Open solution

(A) 11.2 m (B) 8.7 m (C) 9.5 m (D) 10.5 m

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Q112. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

A thin uniform tube is bent into a circle of radius r in the vertical plane. Equal volumes of two immiscible liquids, whose densities are ρ_1 and ρ_2 $(\rho_1 > \rho_2)$, fill half the circle. The angle θ between the radius vector passing through the common interface and the vertical is :

(A) $\theta = \tan^{-1}\pi\left(\frac{\rho_1}{\rho_2}\right)$

(B) $\theta = \tan^{-1}\frac{\pi}{2}\left(\frac{\rho_1}{\rho_2}\right)$

(C) $\theta = \tan^{-1}\left(\frac{\rho_1-\rho_2}{\rho_1+\rho_2}\right)$

(D) $\theta = \tan^{-1}\frac{\pi}{2}\left(\frac{\rho_1+\rho_2}{\rho_1-\rho_2}\right)$

Q113. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

Two tubes of radii r_1 and r_2, and lengths l_1 and l_2 , respectively, are connected in series and a liquid flows through each of them in stream line conditions. P_1 and P_2 are pressure differences across the two tubes.

If P_2 is $4P_1$ and l_2 is $\frac{l_1}{4}$, then the radius r_2 will be equal to :

(A) r_1 (B) $2r_1$ (C) $4r_1$ (D) $\frac{r_1}{2}$

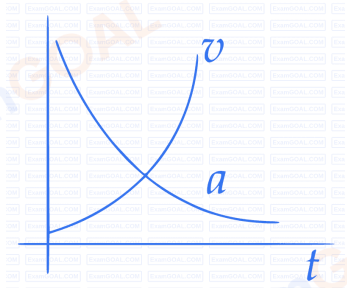
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Q114. JEE Main 2016 (Online) 9th April Morning Slot

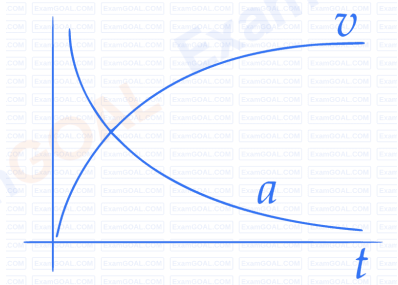
Open solution

Which of the following option correctly describes the variation of the speed *v* and acceleration *'a'* of a point mass falling vertically in a viscous medium that applies a force F = − *kv*, where 'k' is a constant, on the body ? (Graphs are schematic and not drawn to scale)

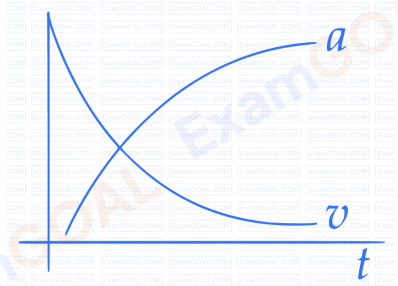
(A)



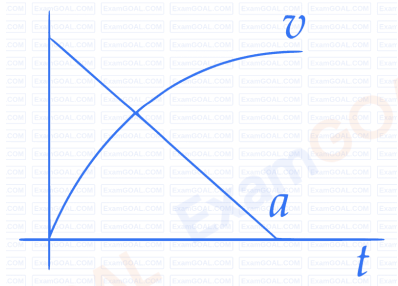
(B)



(C)



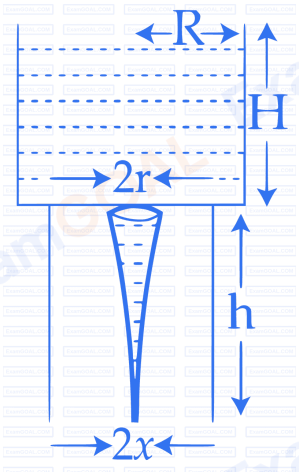
(D)



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Q115. JEE Main 2016 (Online) 9th April Morning Slot

Open solution



Consider a water jar of radius R that has water filled up to height H and is kept on astand of height h (see figure). Through a hole of radius r (r << R) at its bottom, the water leaks out and the stream of water coming down towards the ground has a shape like a funnel as shown in the figure. If the radius of the cross-section of water stream when it hits the ground is x. Then :

(A) $x = r\left(\frac{H}{H+h}\right)$

(B) $x = r\left(\frac{H}{H+h}\right)^{\frac{1}{2}}$

(C) $x = r\left(\frac{H}{H+h}\right)^{\frac{1}{4}}$

(D) $x = r\left(\frac{H}{H+h}\right)^{2}$

Q116. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

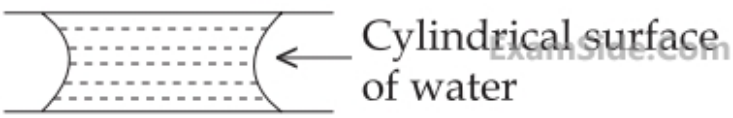
If it takes 5 minutes to fill a 15 litre bucket from a water tap of diameter $\frac{2}{\sqrt{\pi}}$ cm then the Reynolds number for the flow is (density of water = 10^3 kg/m^3 and viscosity of water = 10^{-3} Pa.s) close to :

(A) 5500 (B) 11,000 (C) 550 (D) 1100

Q117. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

If two glass plates have water between them and are separated by very small distance (see figure), it is very difficult to pull them apart. It is because the water in between forms cylindrical surface on the side that gives rise to lower pressure in the water in comparison to atmosphere. If the radius of the cylindrical surface is R and surface tension of water is T then the pressure in water between the plates is lower by :



(A) $\frac{T}{4R}$ (B) $\frac{T}{2R}$ (C) $\frac{4T}{R}$ (D) $\frac{2T}{R}$

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Q118. JEE Main 2014 (Online) 19th April Morning Slot

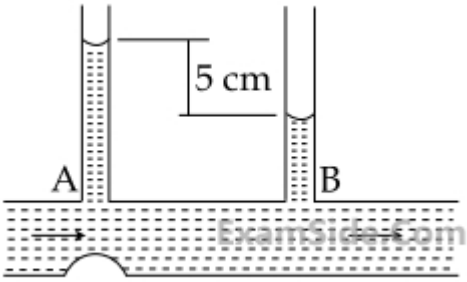
Open solution

The velocity of water in a river is 18 km/hr near the surface. If the river is 5 m deep, find the shearing stress between the horizontal layers of water. The co-efficient of viscosity of water = 10^{-2} poise.

(A) 10^{-1} N/m^2 (B) 10^{-2} N/m^2 (C) 10^{-3} N/m^2 (D) 10^{-4} N/m^2

Q119. JEE Main 2014 (Online) 19th April Morning Slot

Open solution



In the diagram shown, if the difference in the two tubes of the manometer is 5 cm, the cross section of the tube at A and B is 6 mm^2 and 10 mm^2 respectively. The rate at which water flows through the tube (g = 10 ms^{-2})

(A) 7.5 cc/s (B) 8.0 cc/s (C) 10.0 cc/s (D) 12.5 cc/s

Q120. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

A large number of liquid drops each of radius r coalesce to form a single drop of radius R. The energy released in the process is converted into kinetic energy of the big drop so formed. The speed of the big drop is (given surface tension of liquid T, density ρ)

(A) $\sqrt{\frac{T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$ (B) $\sqrt{\frac{2T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$

(C) $\sqrt{\frac{4T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$ (D) $\sqrt{\frac{6T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$

Q121. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

A cylindrical vessel of cross-section A contains water to a height h. There is a hole in the bottom of radius 'a'. The time in which it will be emptied is :

(A) $\frac{2A}{\pi a^2}\sqrt{\frac{h}{g}}$ (B) $\frac{\sqrt{2}A}{\pi a^2}\sqrt{\frac{h}{g}}$

(C) $\frac{2\sqrt{2}A}{\pi a^2}\sqrt{\frac{h}{g}}$ (D) $\frac{A}{\sqrt{2}\,\pi a^2}\sqrt{\frac{h}{g}}$

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Q122. JEE Main 2014 (Online) 12th April Morning Slot

Two soap bubbles coalesce to form a single bubble. If V is the subsequent change in volume of contained air and S the change in total surface area, T is the surface tension and P atmospheric pressure, which of the following relation is **correct** ?

Open solution

(A) 4PV + 3ST = 0 (B) 3PV + 4ST = 0 (C) 2PV + 3ST = 0 (D) 3PV + 2ST = 0

Q123. JEE Main 2014 (Online) 11th April Morning Slot

A tank with a small hole at the bottom has been filled with water and kerosene (specific gravity 0.8). The height of water is 3 m and that of kerosene 2 m. When the hole is opened the velocity of fluid coming out from it is nearly : (take g = 10 ms^{-2} and density of water = 10^3 kg m^{-3})

Open solution

(A) 10.7 ms^{-1} (B) 9.6 ms^{-1} (C) 8.5 ms^{-1} (D) 7.6 ms^{-1}

Q124. JEE Main 2014 (Online) 11th April Morning Slot

An air bubble of radius 0.1 cm is in a liquid having surface tension 0.06 N/m and density 10^3 kg/m^3. The pressure inside the bubble is 1100 Nm^{-2} greater than the atmospheric pressure. At what depth is the bubble below the surface of the liquid ? (g = 9.8 ms^{-2})

Open solution

(A) 0.1 m (B) 0.15 m (C) 0.20 m (D) 0.25 m

Q125. JEE Main 2014 (Online) 9th April Morning Slot

A capillary tube is immersed vertically in water and the height of the water column is x. When this arrangement is taken into a mine of depth d, the height of the water column is y. If R is the radius of earth, the ratio $\frac{x}{y}$ is :

Open solution

(A) $\left(1-\frac{d}{R}\right)$ (B) $\left(1-\frac{2d}{R}\right)$
(C) $\left(\frac{R-d}{R+d}\right)$ (D) $\left(\frac{R+d}{R-d}\right)$

Q126. JEE Main 2014 (Offline)

An open glass tube is immersed in mercury in such a way that a length of $8\ cm$ extends above the mercury level. The open end of the tube is then closed and sealed and the tube is raised vertically up by additional $46\ cm$. What will be length of the air column above mercury in the tube now? (Atmospheric pressure $= 76\ cm$ of Hg)

Open solution

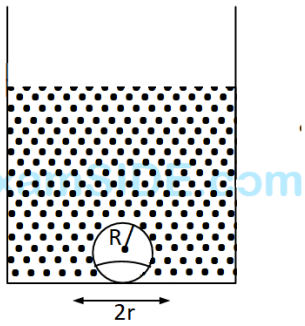
(A) $16\ cm$ (B) $22\ cm$ (C) $38\ cm$ (D) $6\ cm$

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Q127. JEE Main 2014 (Offline)

Open solution

On heating water, bubbles being formed at the bottom of the vessel detach and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If $r << R$ and the surface tension of water is T, value of r just before bubbles detach is: (density of water is ρ_w)



(A) $R^2\sqrt{\frac{\rho_w g}{3T}}$

(B) $R^2\sqrt{\frac{\rho_w g}{6T}}$

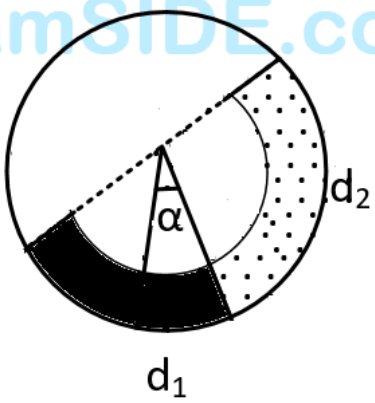
(C) $R^2\sqrt{\frac{\rho_w g}{T}}$

(D) $R^2\sqrt{\frac{2\rho_w g}{3T}}$

Q128. JEE Main 2014 (Offline)

Open solution

There is a circular tube in a vertical plane. Two liquids which do not mix and of densities d_1 and d_2 are filled in the tube. Each liquid subtends 90° angle at center. Radius joining their interface makes an angle α with vertical. Ratio $\frac{d_1}{d_2}$ is :



(A) $\frac{1+\sin\alpha}{1-\sin\alpha}$

(B) $\frac{1+\cos\alpha}{1-\cos\alpha}$

(C) $\frac{1+\tan\alpha}{1-\tan\alpha}$

(D) $\frac{1+\sin\alpha}{1-\cos\alpha}$

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Q129. JEE Main 2013 (Online) 25th April Morning Slot

In an experiment, a small steel ball falls through a liquid at a constant speed of 10 cm/s. If the steel ball is pulled upward with a force equal to twice its effective weight, how fast will it move upward?

(A) 5 cm/s (B) Zero (C) 10 cm/s (D) 20 cm/s

Open solution

Q130. JEE Main 2013 (Online) 23th April Morning Slot

Wax is coated on the inner wall of a capillary tube and the tube is then dipped in water. Then, compared to the unwaxed capillary, the angle of contact θ and the height h upto which water rises change. These changes are :

(A) θ increases and h also increases

(B) θ decreases and h also decreases

(C) θ increases and h decreases

(D) θ decreases and h increases

Open solution

Q131. JEE Main 2013 (Online) 22th April Morning Slot

Air of density 1.2 kg m^{-3} is blowing across the horizontal wings of an aeroplane in such a way that its speeds above and below the wings are 150 ms^{-1} and 100 ms^{-1}, respectively. The pressure difference between the upper and lower sides of the wings, is :

(A) 60 Nm^{-2} (B) 180 Nm^{-2} (C) 7500 Nm^{-2} (D) 12500 Nm^{-2}

Open solution

Q132. JEE Main 2013 (Online) 9th April Morning Slot

This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement-1: A capillary is dipped in a liquid and liquid rises to a height h in it. As the temperature of the liquid is raised, the height h increases (if the density of the liquid and the angle of contact remain the same).

Statement-2: Surface tension of a liquid decreases with the rise in its temperature.

(A) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation for Statement-1.

(B) Statement-1 is false, Statement-2 is true.

(C) Statement-1 is true, Statement-2 is false.

(D) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation for Statement-1.

Open solution

Q133. JEE Main 2013 (Offline)

A uniform cylinder of length L and mass M having cross-sectional area A is suspended, with its length vertical, from a fixed point by a mass-less spring such that it is half submerged in a liquid of density σ at equilibrium position. The extension x_0 of the spring when it is in equilibrium is:

(A) $\frac{Mg}{k}$

(B) $\frac{Mg}{k}\left(1-\frac{LA\sigma}{M}\right)$

(C) $\frac{Mg}{k}\left(1-\frac{LA\sigma}{2M}\right)$

(D) $\frac{Mg}{k}\left(1+\frac{LA\sigma}{M}\right)$

Open solution

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Q134. JEE Main 2013 (Offline)

Assume that a drop of liquid evaporates by decreases in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible ? The surface tension is T, density of liquid is ρ and L is its latent heat of vaporization.

Open solution

(A) $\rho L/T$ (B) $\sqrt{T/\rho L}$ (C) $T/\rho L$ (D) $2T/\rho L$

Q135. AIEEE 2012 (Online) 26th May

The terminal velocity of a small sphere of radius a in a viscous liquid is proportional to

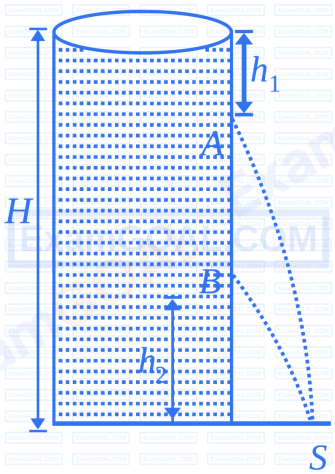
Open solution

(A) a^2 (B) a^3 (C) a (D) a^{-1}

Q136. AIEEE 2012 (Online) 26th May

In a cylindrical water tank, there are two small holes A and B on the wall at a depth of h_1 from the surface of water and at a height of h_2 from the bottom of water tank. Surface of water is at height H from the bottom of water tank. Water coming out from both holes strikes the ground at the same point S. Find the ratio of h_1 and h_2

Open solution



(A) Depends on H (B) $1:1$ (C) $2:2$ (D) $1:2$

Q137. AIEEE 2012 (Online) 19th May

A large number of droplets, each of radius, r coalesce to form a bigger drop of radius, R. An engineer designs a machine so that the energy released in this process is converted into the kinetic energy of the drop. Velocity of the drop is (T = surface tension, ρ = density)

Open solution

(A) $\left[\frac{T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)\right]^{1/2}$ (B) $\left[\frac{6T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)\right]^{1/2}$

(C) $\left[\frac{3T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)\right]^{1/2}$ (D) $\left[\frac{2T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)\right]^{1/2}$

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Q138. AIEEE 2012 (Online) 12th May

Water is flowing through a horizontal tube having cross-sectional areas of its two ends being A and A' such that the ratio A/A' is 5. If the pressure difference of water between the two ends is $3 \times 10^5\ \mathrm{N\,m^{-2}}$, the velocity of water with which it enters the tube will be (neglect gravity effects)

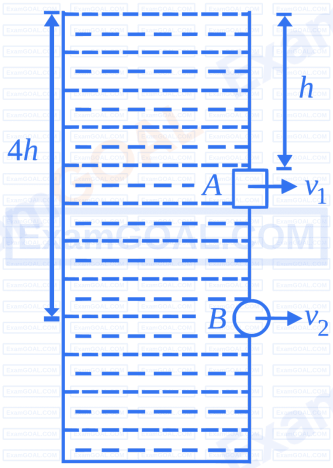
Open solution

(A) $5\ \mathrm{m\,s^{-1}}$ (B) $10\ \mathrm{m\,s^{-1}}$ (C) $25\ \mathrm{m\,s^{-1}}$ (D) $50\sqrt{10}\ \mathrm{m\,s^{-1}}$

Q139. AIEEE 2012 (Online) 7th May

A square hole of side length ℓ is made at a depth of h and a circular hole of radius r is made at a depth of $4h$ from the surface of water in a water tank kept on a horizontal surface. If $\ell \ll h, r \ll h$ and the rate of water flow from the holes is the same, then r is equal to

Open solution

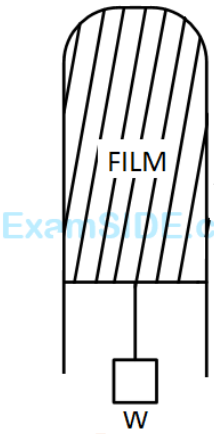


(A) $\frac{\ell}{\sqrt{2\pi}}$ (B) $\frac{\ell}{\sqrt{3\pi}}$ (C) $\frac{\ell}{3\pi}$ (D) $\frac{\ell}{2\pi}$

Q140. AIEEE 2012

A thin liquid film formed between a U-shaped wire and a light slider supports a weight of $1.5 \times 10^{-2}\ N$ (see figure). The length of the slider is $30\ cm$ and its weight negligible. The surface tension of the liquid film is

Open solution



(A) $0.0125\ Nm^{-1}$ (B) $0.1\ Nm^{-1}$ (C) $0.05\ Nm^{-1}$ (D) $0.025\ Nm^{-1}$

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Q141. AIEEE 2011

Work done in increasing the size of a soap bubble from a radius of $3\ cm$ to $5\ cm$ is nearly (Surface tension of soap solution $= 0.03 Nm^{-1}$,

(A) $0.2\pi mJ$ (B) $2\pi mJ$ (C) $0.4\pi mJ$ (D) $4\pi mJ$

Open solution

Q142. AIEEE 2011

Water is flowing continuously from a tap having an internal diameter $8 \times 10^{-3}\ m$. The water velocity as it leaves the tap is $0.4\ ms^{-1}$. The diameter of the water stream at a distance $2 \times 10^{-1}\ m$ below the tap is close to :

(A) $7.5 \times 10^{-3} m$ (B) $9.6 \times 10^{-3} m$ (C) $3.6 \times 10^{-3} m$ (D) $5.0 \times 10^{-3} m$

Open solution

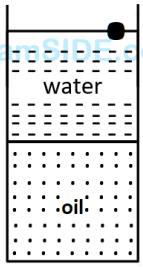
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Q143. AIEEE 2010

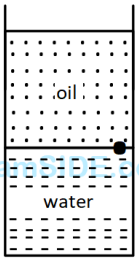
Open solution

A ball is made of a material of density ρ where $\rho_{oil} < \rho < \rho_{water}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position?

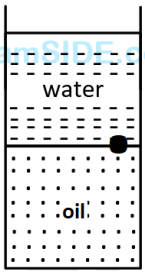
(A)



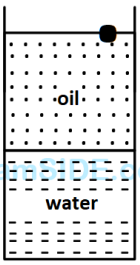
(B)



(C)



(D)



Q144. AIEEE 2008

Open solution

A spherical solid ball of volume V is made of a material of density ρ_1. It is falling through a liquid of density ρ_2 $(\rho_2 < \rho_1)$. Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v, i.e., $F_{viscous} = -kv^2$ $(k > 0)$. The terminal speed of the ball is

(A) $\sqrt{\frac{Vg(\rho_1-\rho_2)}{k}}$

(B) $\frac{Vg\rho_1}{k}$

(C) $\sqrt{\frac{Vg\rho_1}{k}}$

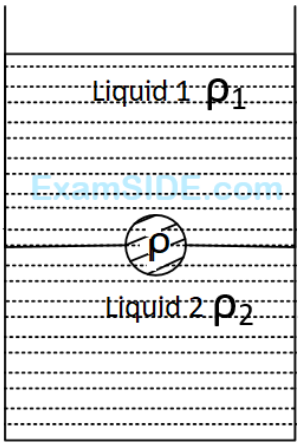
(D) $\frac{Vg(\rho_1-\rho_2)}{k}$

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Q145. AIEEE 2008

A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and ρ_2 respectively. A solid ball, made of a material of density ρ_3, is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1 , ρ_1 and ρ_3 ?

Open solution



(A) $\rho_3 < \rho_1 < \rho_2$

(B) $\rho_1 > \rho_3 > \rho_2$

(C) $\rho_1 < \rho_2 < \rho_3$

(D) $\rho_1 < \rho_3 < \rho_2$

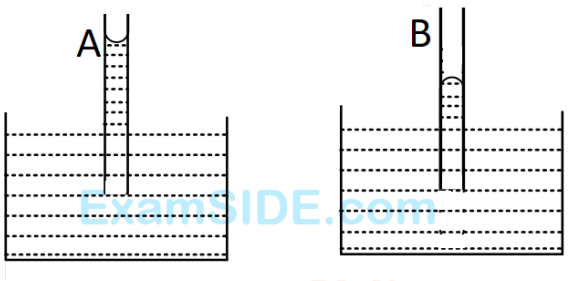
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Q146. AIEEE 2008

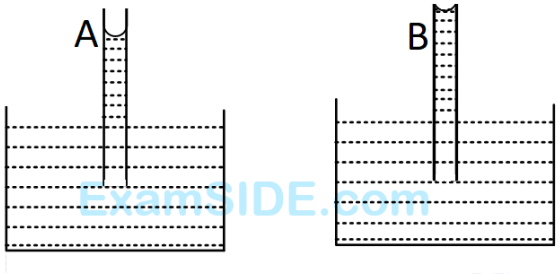
A capillary tube (A) is dipped in water. Another identical tube (B) is dipped in a soap-water solution. Which of the following shows the relative nature of the liquid columns in the two tubes?

Open solution

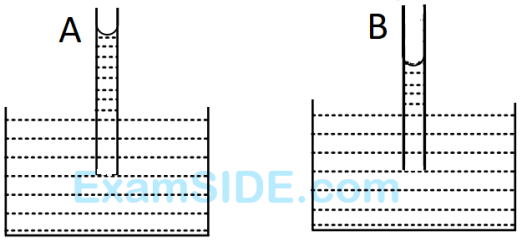
(A)



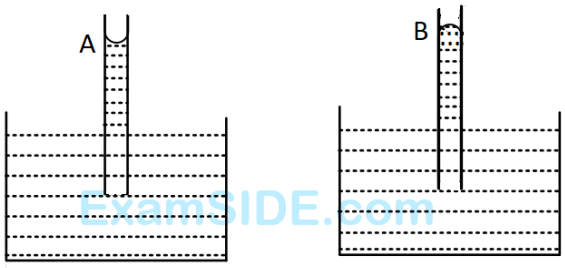
(B)



(C)



(D)



Q147. AIEEE 2006

If the terminal speed of a sphere of gold (density $= 19.5\ kg/m^3$) is $0.2\ m/s$ in a viscous liquid (density $= 1.5\ kg/m^3$, find the terminal speed of a sphere of silver (density $= 10.5\ kg/m^3$) of the same size in the same liquid

Open solution

(A) $0.4\ m/s$ (B) $0.133\ m/s$ (C) $0.1\ m/s$ (D) $0.2\ m/s$

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Q148. AIEEE 2005

A $20\ cm$ long capillary tube is dipped in water. The water rises up to $8\ cm$. If the entire arrangement is put in a freely falling elevator the length of water column in the capillary tube will be

(A) $10\ cm$ (B) $8\ cm$ (C) $20\ cm$ (D) $4\ cm$

Open solution

Q149. AIEEE 2004

Spherical balls of radius R are falling in a viscous fluid of viscosity η with a velocity v. The retarding viscous force acting on the spherical ball is

(A) inversely proportional to both radius R and velocity v

(B) directly proportional to both radius R and velocity v

(C) directly proportional to R but inversely proportional to v

(D) inversely proportional to R, but directly proportional to velocity v

Open solution

Q150. AIEEE 2004

If two soap bubbles of different radii are connected by a tube

(A) air flows from the smaller bubble to the bigger

(B) air flows from bigger bubble to the smaller bubble till the sizes are interchanged

(C) air flows from the bigger bubble to the smaller bubble till the sizes become equal

(D) there is no flow of air.

Open solution

Q151. AIEEE 2002

A cylinder of height $20\ m$ is completely filled with water. The velocity of efflux of water (in ms^{-1}) through a small hole on the side wall of the cylinder near its bottom is

(A) 10 (B) 20 (C) 25.5 (D) 5

Open solution

Numerical 46 questions

Q152. JEE Main 2026 (Online) 4th April Morning Shift

The surface tension of a soap solution is $3.5 \times 10^{-2}\ \mathrm{N/m}$. The work required to increase the radius of a soap bubble from 1 cm to 2 cm is $\alpha \times 10^{-6}$ J. The value of α is ____.

$(\pi = 22/7)$

Open solution

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Q153. JEE Main 2026 (Online) 2nd April Morning Shift

A tub is filled with water and a wooden cube $10\text{ cm} \times 10\text{ cm} \times 10\text{ cm}$ is placed in the water. The wooden cube is found to float on the water with a part of it submerged in water. When a metal coin is placed on the wooden cube, the submerged part is increased by 3.87 cm. The mass of the metal coin is _____ gram.

(Take water density as 1 g/cm^3 and density of wood as 0.4 g/cm^3)

Open solution

Q154. JEE Main 2026 (Online) 24th January Evening Shift

A soap bubble of surface tension 0.04 N/m is blown to a diameter of 7 cm . If $(15000 - x)\mu\text{J}$ of work is done in blowing it further to make its diameter 14 cm , then the value of x is ____ .

$(\pi = 22/7)$

Open solution

Q155. JEE Main 2026 (Online) 24th January Morning Shift

Sixty four rain drops of radius 1 mm each falling down with a terminal velocity of 10 cm/s coalesce to form a bigger drop. The terminal velocity of bigger drop is

____ cm/s.

Open solution

Q156. JEE Main 2026 (Online) 21st January Evening Shift

The terminal velocity of a metallic ball of radius 6 mm in a viscous fluid is 20 cm/s. The terminal velocity of another ball of same material and having radius 3 mm in the same fluid will be ____ cm/s.

Open solution

Q157. JEE Main 2025 (Online) 8th April Evening Shift

A cube having a side of 10 cm with unknown mass and 200 gm mass were hung at two ends of a uniform rigid rod of 27 cm long. The rod along with masses was placed on a wedge keeping the distance between wedge point and 200 gm weight as 25 cm. Initially the masses were not at balance. A beaker is placed beneath the unknown mass and water is added slowly to it. At given point the masses were in balance and half volume of the unknown mass was inside the water.

(Take the density of unknown mass is more than that of the water, the mass did not absorb water and water density is 1 gm/cm^3.)

The unknown mass is ____ kg.

Open solution

Q158. JEE Main 2025 (Online) 3rd April Evening Shift

The excess pressure inside a soap bubble A in air is half the excess pressure inside another soap bubble B in air. If the volume of the bubble A is n times the volume of the bubble B, then, the value of n is_____.

Open solution

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Q159. JEE Main 2025 (Online) 2nd April Morning Shift

A vessel with square cross-section and height of 6 m is vertically partitioned. A small window of $100\ \mathrm{cm}^2$ with hinged door is fitted at a depth of 3 m in the partition wall. One part of the vessel is filled completely with water and the other side is filled with the liquid having density $1.5 \times 10^3\ \mathrm{kg/m^3}$. What force one needs to apply on the hinged door so that it does not get opened ?

(Acceleration due to gravity $= 10\ \mathrm{m/s^2}$)

Open solution

Q160. JEE Main 2025 (Online) 29th January Morning Shift

In a hydraulic lift, the surface area of the input piston is 6 cm^2 and that of the output piston is 1500 cm^2. If 100 N force is applied to the input piston to raise the output piston by 20 cm, then the work done is _____ kJ.

Open solution

Q161. JEE Main 2025 (Online) 23rd January Evening Shift

An air bubble of radius 1.0 mm is observed at a depth 20 cm below the free surface of a liquid having surface tension $0.095\ \mathrm{J/m^2}$ and density $10^3\ \mathrm{kg/m^3}$. The difference between pressure inside the bubble and atmospheric pressure is _____ $\mathrm{N/m^2}$. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

Open solution

Q162. JEE Main 2025 (Online) 22nd January Morning Shift

Two soap bubbles of radius 2 cm and 4 cm , respectively, are in contact with each other. The radius of curvature of the common surface, in cm , is _____.

Open solution

Q163. JEE Main 2024 (Online) 8th April Evening Shift

Small water droplets of radius $0.01\ \mathrm{mm}$ are formed in the upper atmosphere and falling with a terminal velocity of $10\ \mathrm{cm/s}$. Due to condensation, if 8 such droplets are coalesced and formed a larger drop, the new terminal velocity will be _____ $\mathrm{cm/s}$.

Open solution

Q164. JEE Main 2024 (Online) 8th April Morning Shift

A liquid column of height $0.04\ \mathrm{cm}$ balances excess pressure of a soap bubble of certain radius. If density of liquid is $8 \times 10^3\ \mathrm{kg\ m^{-3}}$ and surface tension of soap solution is $0.28\ \mathrm{Nm^{-1}}$, then diameter of the soap bubble is _____ cm. (if $\mathrm{g} = 10\ \mathrm{m\ s^{-2}}$)

Open solution

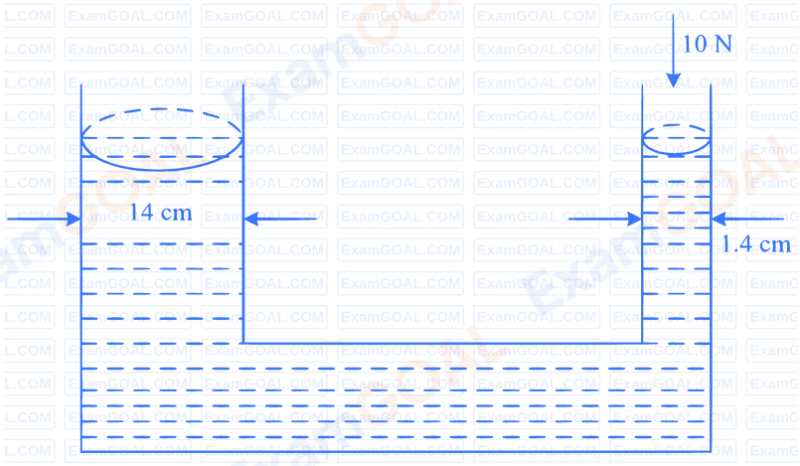
Q165. JEE Main 2024 (Online) 6th April Morning Shift

A big drop is formed by coalescing 1000 small droplets of water. The ratio of surface energy of 1000 droplets to that of energy of big drop is $\frac{10}{x}$. The value of x is _____.

Open solution

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Q166. JEE Main 2024 (Online) 5th April Evening Shift



Open solution

A hydraulic press containing water has two arms with diameters as mentioned in the figure. A force of $10\ \mathrm{N}$ is applied on the surface of water in the thinner arm. The force required to be applied on the surface of water in the thicker arm to maintain equilibrium of water is _____ N.

Q167. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

Mercury is filled in a tube of radius $2\ \mathrm{cm}$ up to a height of $30\ \mathrm{cm}$. The force exerted by mercury on the bottom of the tube is _____ N.

(Given, atmospheric pressure $= 10^5\ \mathrm{Nm}^{-2}$, density of mercury $= 1.36 \times 10^4\ \mathrm{kg\,m}^{-3}$, $\mathrm{g} = 10\ \mathrm{m\,s}^{-2}, \pi = \frac{22}{7}$)

Q168. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

A soap bubble is blown to a diameter of $7\ \mathrm{cm}$. $36960\ \mathrm{erg}$ of work is done in blowing it further. If surface tension of soap solution is 40 dyne/cm then the new radius is _____ cm. Take $(\pi = \frac{22}{7})$.

Q169. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

A plane is in level flight at constant speed and each of its two wings has an area of $40\ \mathrm{m}^2$. If the speed of the air is $180\ \mathrm{km/h}$ over the lower wing surface and $252\ \mathrm{km/h}$ over the upper wing surface, the mass of the plane is _____ kg.

(Take air density to be $1\ \mathrm{kg\,m}^{-3}$ and $\mathrm{g} = 10\ \mathrm{ms}^{-2}$)

Q170. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

A big drop is formed by coalescing 1000 small identical drops of water. If E_1 be the total surface energy of 1000 small drops of water and E_2 be the surface energy of single big drop of water, then $E_1 : E_2$ is $x : 1$ where $x =$ _____.

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Q171. JEE Main 2024 (Online) 29th January Morning Shift

In a test experiment on a model aeroplane in wind tunnel, the flow speeds on the upper and lower surfaces of the wings are $70\ \mathrm{ms}^{-1}$ and $65\ \mathrm{ms}^{-1}$ respectively. If the wing area is $2\ \mathrm{m}^2$, the lift of the wing is _____ N.

(Given density of air $= 1.2\ \mathrm{kg\ m}^{-3}$)

Open solution

Q172. JEE Main 2024 (Online) 27th January Evening Shift

The reading of pressure metre attached with a closed pipe is $4.5 \times 10^4\ \mathrm{N/m}^2$. On opening the valve, water starts flowing and the reading of pressure metre falls to $2.0 \times 10^4\ \mathrm{N/m}^2$. The velocity of water is found to be $\sqrt{V}\ \mathrm{m/s}$. The value of V is _____.

Open solution

Q173. JEE Main 2023 (Online) 15th April Morning Shift

There is an air bubble of radius $1.0\ \mathrm{mm}$ in a liquid of surface tension $0.075\ \mathrm{Nm}^{-1}$ and density $1000\ \mathrm{kg\ m}^{-3}$ at a depth of $10\ \mathrm{cm}$ below the free surface. The amount by which the pressure inside the bubble is greater than the atmospheric pressure is _____ $\mathrm{Pa}\ \left(\mathrm{g} = 10\ \mathrm{ms}^{-2}\right)$

Open solution

Q174. JEE Main 2023 (Online) 12th April Morning Shift

Glycerin of density $1.25 \times 10^3\ \mathrm{kg\ m}^{-3}$ is flowing through the conical section of pipe. The area of cross-section of the pipe at its ends are $10\ \mathrm{cm}^2$ and $5\ \mathrm{cm}^2$ and pressure drop across its length is $3\ \mathrm{Nm}^{-2}$. The rate of flow of glycerin through the pipe is $x \times 10^{-5}\ \mathrm{m}^3\ \mathrm{s}^{-1}$. The value of x is _____.

Open solution

Q175. JEE Main 2023 (Online) 11th April Evening Shift

The surface tension of soap solution is $3.5 \times 10^{-2}\ \mathrm{Nm}^{-1}$. The amount of work done required to increase the radius of soap bubble from $10\ \mathrm{cm}$ to $20\ \mathrm{cm}$ is _____ $\times 10^{-4}$ J.

$(\text{take}\ \pi = 22/7)$

Open solution

Q176. JEE Main 2023 (Online) 10th April Evening Shift

Figure below shows a liquid being pushed out of the tube by a piston having area of cross section $2.0\ \mathrm{cm}^2$. The area of cross section at the outlet is $10\ \mathrm{mm}^2$. If the piston is pushed at a speed of $4\ \mathrm{cm\ s}^{-1}$, the speed of outgoing fluid is _____ cms^{-1}



Open solution

Q177. JEE Main 2023 (Online) 8th April Morning Shift

An air bubble of diameter $6\ \mathrm{mm}$ rises steadily through a solution of density $1750\ \mathrm{kg/m}^3$ at the rate of $0.35\ \mathrm{cm/s}$. The co-efficient of viscosity of the solution (neglect density of air) is _____ Pas (given, $\mathrm{g} = 10\ \mathrm{ms}^{-2}$).

Open solution

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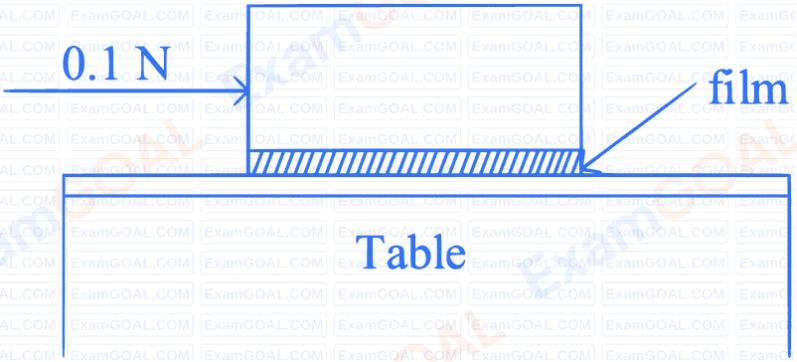
Q178. JEE Main 2023 (Online) 1st February Evening Shift

The surface of water in a water tank of cross section area $750\ \mathrm{cm}^2$ on the top of a house is $h\ \mathrm{m}$ above the tap level. The speed of water coming out through the tap of cross section area $500\ \mathrm{mm}^2$ is $30\ \mathrm{cm/s}$. At that instant, $\frac{dh}{dt}$ is $x \times 10^{-3}\ \mathrm{m/s}$. The value of x will be _____.

Open solution

Q179. JEE Main 2023 (Online) 29th January Evening Shift

A metal block of base area 0.20 m^2 is placed on a table, as shown in figure. A liquid film of thickness 0.25 mm is inserted between the block and the table. The block is pushed by a horizontal force of 0.1 N and moves with a constant speed. If the viscosity of the liquid is $5.0 \times 10^{-3}\ \mathrm{Pl}$, the speed of block is _____ $\times 10^{-3}$ m/s.



Open solution

Q180. JEE Main 2023 (Online) 25th January Evening Shift

A spherical drop of liquid splits into 1000 identical spherical drops. If u_i is the surface energy of the original drop and u_f is the total surface energy of the resulting drops, then (ignoring evaporation), $\frac{u_f}{u_i} = \left(\frac{10}{x}\right)$. Then value of x is _____ :

Open solution

Q181. JEE Main 2023 (Online) 24th January Evening Shift

A Spherical ball of radius 1mm and density 10.5 g/cc is dropped in glycerine of coefficient of viscosity 9.8 poise and density 1.5 g/cc. Viscous force on the ball when it attains constant velocity is 3696×10^{-x} N. The value of x is _____.

(Given, g = 9.8 m/s^2 and $\pi = \frac{22}{7}$)

Open solution

Q182. JEE Main 2022 (Online) 30th July Evening Shift

A water drop of radius $1\ \mathrm{cm}$ is broken into eight equal droplets. Surface tension of water is $0.075\ \mathrm{N\,m^{-1}}$. The gain in surface energy is _____ $\times 10^{-7}$ J. (Take $\pi = 3.14$)

Open solution

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Q183. JEE Main 2022 (Online) 29th July Evening Shift

A tube of length $50\ \text{cm}$ is filled completely with an incompressible liquid of mass $250\ \text{g}$ and closed at both ends. The tube is then rotated in horizontal plane about one of its ends with a uniform angular velocity $x\sqrt{F}\ \text{rad s}^{-1}$. If F be the force exerted by the liquid at the other end then the value of x will be _____.

Open solution

Q184. JEE Main 2022 (Online) 29th July Evening Shift

The velocity of a small ball of mass $0.3\ \text{g}$ and density $8\ \text{g/cc}$ when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $1.3\ \text{g/cc}$, then the value of viscous force acting on the ball will be $x \times 10^{-4}\ \text{N}$. The value of x is _____. [use $\text{g} = 10\ \text{m/s}^2$]

Open solution

Q185. JEE Main 2022 (Online) 28th July Morning Shift

The diameter of an air bubble which was initially $2\ \text{mm}$, rises steadily through a solution of density $1750\ \text{kg m}^{-3}$ at the rate of $0.35\ \text{cms}^{-1}$. The coefficient of viscosity of the solution is _____ poise (in nearest integer). (the density of air is negligible).

Open solution

Q186. JEE Main 2022 (Online) 27th July Evening Shift

A spherical soap bubble of radius 3 cm is formed inside another spherical soap bubble of radius 6 cm. If the internal pressure of the smaller bubble of radius 3 cm in the above system is equal to the internal pressure of another single soap bubble of radius r cm. The value of r is _____.

Open solution

Q187. JEE Main 2022 (Online) 30th June Morning Shift

The excess pressure inside a liquid drop is 500 Nm^{-2}. If the radius of the drop is 2 mm, the surface tension of liquid is x $\times$ 10^{-3} Nm^{-1}. The value of x is _____.

Open solution

Q188. JEE Main 2022 (Online) 29th June Evening Shift

A small spherical ball of radius 0.1 mm and density 10^4 kg m^{-3} falls freely under gravity through a distance h before entering a tank of water. If, after entering the water the velocity of ball does not change and it continue to fall with same constant velocity inside water, then the value of h will be _____ m.

(Given g = 10 ms^{-2}, viscosity of water = 1.0 $\times$ 10^{-5} N-sm^{-2}).

Open solution

Q189. JEE Main 2022 (Online) 28th June Evening Shift

A liquid of density 750 kgm^{-3} flows smoothly through a horizontal pipe that tapers in cross-sectional area from A_1 = 1.2 $\times$ 10^{-2} m^2 to $A_2 = \frac{A_1}{2}$. The pressure difference between the wide and narrow sections of the pipe is 4500 Pa. The rate of flow of liquid is _____ $\times$ 10^{-3} m^3s^{-1}.

Open solution

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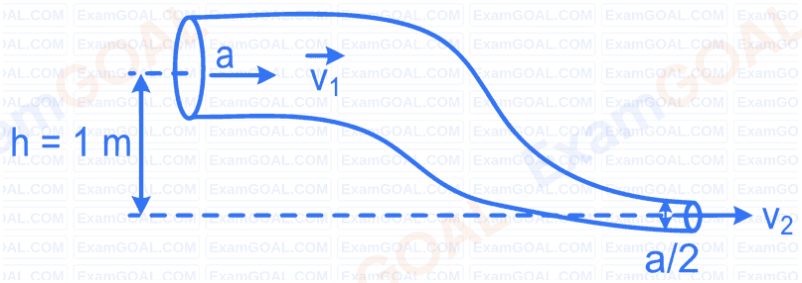
Q190. JEE Main 2022 (Online) 27th June Morning Shift

The area of cross-section of a large tank is 0.5 m^2. It has a narrow opening near the bottom having area of cross-section 1 cm^2. A load of 25 kg is applied on the water at the top in the tank. Neglecting the speed of water in the tank, the velocity of the water, coming out of the opening at the time when the height of water level in the tank is 40 cm above the bottom, will be _____ cms $^{-1}$. [Take g = 10 ms^{-2}]

Open solution

Q191. JEE Main 2022 (Online) 26th June Morning Shift

An ideal fluid of density 800 kgm^{-3}, flows smoothly through a bent pipe (as shown in figure) that tapers in cross-sectional area from a to $\frac{a}{2}$. The pressure difference between the wide and narrow sections of pipe is 4100 Pa. At wider section, the velocity of fluid is $\frac{\sqrt{x}}{6}$ ms^{-1} for x = _____. (Given g = 10 ms^{-2})



Open solution

Q192. JEE Main 2022 (Online) 25th June Morning Shift

The velocity of upper layer of water in a river is 36 kmh^{-1}. Shearing stress between horizontal layers of water is 10^{-3} Nm^{-2}. Depth of the river is _____ m. (Co-efficient of viscosity of water is 10^{-2} Pa.s)

Open solution

Q193. JEE Main 2021 (Online) 26th August Morning Shift

A soap bubble of radius 3 cm is formed inside the another soap bubble of radius 6 cm. The radius of an equivalent soap bubble which has the same excess pressure as inside the smaller bubble with respect to the atmospheric pressure is ..... cm.

Open solution

Q194. JEE Main 2021 (Online) 27th July Evening Shift

The water is filled upto height of 12 m in a tank having vertical sidewalls. A hole is made in one of the walls at a depth 'h' below the water level. The value of 'h' for which the emerging stream of water strikes the ground at the maximum range is _____ m.

Open solution

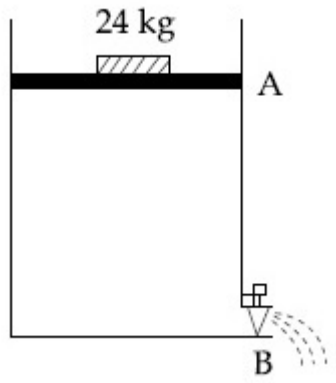
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Q195. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

Consider a water tank as shown in the figure. It's cross-sectional area is 0.4 m^2. The tank has an opening B near the bottom whose cross-section area is 1 cm^2. A load of 24 kg is applied on the water at the top when the height of the water level is 40 cm above the bottom, the velocity of water coming out the opening B is v ms^{-1}.

The value of v, to the nearest integer, is _____. [Take value of g to be 10 ms^{-2}]



Q196. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

A hydraulic press can lift 100 kg when a mass 'm' is placed on the smaller piston. It can lift _____ kg when the diameter of the larger piston is increased by 4 times and that of the smaller piston is decreased by 4 times keeping the same mass 'm' on the smaller piston.

Q197. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

When a long glass capillary tube of radius 0.015 cm is dipped in a liquid, the liquid rises to a height of 15 cm within it. If the contact angle between the liquid and glass to close to 0°, the surface tension of the liquid, in milliNewton m−1, is [$\rho_{(liquid)}$ = 900 kgm^{-3}, g = 10 ms^{-2}]
(Give answer in closest integer) _____.

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **D**	3. **B**	4. **B**	5. **B**
6. **C**	7. **B**	8. **B**	9. **D**	10. **D**
11. **D**	12. **B**	13. **A**	14. **B**	15. **D**
16. **A**	17. **D**	18. **C**	19. **D**	20. **D**
21. **C**	22. **B**	23. **C**	24. **B**	25. **B**
26. **D**	27. **B**	28. **D**	29. **D**	30. **C**
31. **B**	32. **A**	33. **D**	34. **D**	35. **D**
36. **B**	37. **A**	38. **B**	39. **D**	40. **A**
41. **B**	42. **D**	43. **B**	44. **B**	45. **C**
46. **B**	47. **B**	48. **B**	49. **A**	50. **D**
51. **D**	52. **D**	53. **D**	54. **A**	55. **D**
56. **A**	57. **A**	58. **D**	59. **A**	60. **A**
61. **D**	62. **C**	63. **B**	64. **C**	65. **A**
66. **B**	67. **D**	68. **C**	69. **A**	70. **B**
71. **C**	72. **C**	73. **B**	74. **B**	75. **C**
76. **C**	77. **C**	78. **A**	79. **D**	80. **A**
81. **B**	82. **B**	83. **B**	84. **B**	85. **D**
86. **C**	87. **D**	88. **B**	89. **B**	90. **C**
91. **D**	92. **D**	93. **C**	94. **B**	95. **C**
96. **B**	97. **D**	98. **C**	99. **C**	100. **B**
101. **A**	102. **C**	103. **D**	104. **B**	105. **B**
106. **A**	107. **A**	108. **B**	109. **B**	110. **B**
111. **C**	112. **C**	113. **D**	114. **B**	115. **C**
116. **A**	117. *Scan QR*	118. **C**	119. **A**	120. **D**
121. **B**	122. **B**	123. **B**	124. **A**	125. **A**
126. **A**	127. **D**	128. **C**	129. **C**	130. **C**
131. **C**	132. **B**	133. **C**	134. **D**	135. **A**
136. **B**	137. **B**	138. **A**	139. **A**	140. **D**
141. **C**	142. **C**	143. **B**	144. **A**	145. **D**
146. **C**	147. **C**	148. **C**	149. **B**	150. **A**

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151. **B**

Numerical

152. **264**	153. **387**	154. **11304**	155. **160**	156. **5**
157. **3**	158. **8**	159. **150**	160. **5**	161. **2190**
162. **4**	163. **40**	164. **7**	165. **1**	166. **1000**
167. **177**	168. **7**	169. **9600**	170. **10**	171. **810**
172. **50**	173. **1150**	174. **4**	175. **264**	176. **80**
177. **10**	178. **2**	179. **25**	180. **1**	181. **7**
182. **942**	183. **4**	184. **25**	185. **11**	186. **2**
187. **500**	188. **20**	189. **24**	190. **300**	191. **363**
192. **100**	193. **2**	194. **6**	195. **3**	196. **25600**
197. **101**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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